

A review of the nutrition and feeding management of farmed tilapia throughout the culture cycle

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Abstract

Tilapia is the second most farmed fish group worldwide and over the past decade has quadrupled in production, largely due to their many characteristics conducive to aquaculture conditions as well as to the high marketability and relatively stable market prices. To keep pace with this rapid expansion, tilapia aquaculture will rely increasingly on more sustainable practices, but this must not be at the expense of decreased productivity, increased cost or compromised product quality. Since feed constitutes the major operating cost of tilapia farming, understanding their nutritional requirements, factors that may influence these requirements and implementing the most appropriate feeding management strategy is essential for the continued sustainability and scalability of the global tilapia industry. This review addressed these concerns by providing an overview of published nutrient requirement data and offered practical recommendations to the nutrition and feeding management of farmed tilapia, from broodstock to finishing feeds. The first section of this review extensively discusses the dietary protein/amino acids, lipids/fatty acids, carbohydrates, vitamin and mineral requirements of tilapia. Alternatives to marine-based ingredients, which are used increasingly in tilapia feeds, are also discussed. This is followed by discussing practical feed management aspects at the commercial farm level including feed formulations, feed types and feeding rates/frequency throughout the culture cycle. Recent research indicating significant differences in nutrient requirements and utilization efficiencies of improved tilapia strains versus non-improved strains are highlighted. This review aims to provide a comprehensive overview of tilapia nutrition and thus contributes to the continued global expansion of tilapia farming and the demand for their products.

Key words: feed formulation, feed ingredient, feed management, nutrient requirement, tilapia.

Introduction

It is well known that aquaculture is the fastest growing food production sector in the world and produces a variety of products, thus reducing the pressure on the capture fisheries industry which has been on a plateau for many decades. To meet the increasing worldwide demand for protein sources, perhaps one of the most significant fish species bridging this gap are tilapia. Tilapia are a group of freshwater omnivorous cichlids that are native to Africa and subsequently have been introduced, either deliberately or accidentally, throughout the world (Eknath & Hulata 2009). The first introduction of tilapia to Asia was the Mozambique tilapia, *Oreochromis mossambicus*, in the 1940s, although this has been replaced largely by Nile tilapia, *O. niloticus*, due to their better growth performance in ponds. Initially, these fish, sometimes referred to as 'aquatic chickens', were cultured extensively within inland ponds. However, due to many beneficial characteristics that include their tolerance to crowding, high marketability, relative ease of captive spawning year-round, high disease resistance, success with polyculture and the ability to accept lower cost diets from terrestrial-based ingredients, over 100 tropical and sub-tropical countries now culture tilapia in various culture systems (El-Sayed 1999; Shiao 2002; Food & Agriculture Organization of the United Nations (FAO) 2011a).

Currently, behind carp, tilapia is the second most farmed fish in the world yielding over 3 million metric tons y^{-1} in 2009 (Food & Agriculture Organization of the United

Nations (FAO) 2011a). Of the various species of tilapia, *Oreochromis* spp., *Tilapia* spp. and *Sarotherodon* spp. are the most commercially important, with Nile tilapia being the most commonly farmed tilapia species with an aquaculture production of 2.5 million metric tons y^{-1} in 2009 (Food & Agriculture Organization of the United Nations (FAO) 2011a) (Fig. 1). By far, the vast majority of aquaculture production of tilapia occurs in China at over 1.2 million metric tons in 2009, most of which is Nile tilapia, with Indonesia being the second highest producer at over 250 000 metric tons in 2009 culturing both Nile tilapia and other tilapia species (Food & Agriculture Organization of the United Nations (FAO) 2011a) (Fig. 2a,b). While the USA contributed only 9979 metric tons of farmed tilapia (<0.5% of worldwide production), their imports make up 91.8% of global exports, although there are indications that Europe will increasingly adopt tilapia as an alternative to traditional fish species. Indeed the worldwide trade of tilapia has risen from US\$1.7 billion in 2000 to US\$5.0 billion in 2010 and this trend is likely to continue (Food & Agriculture Organization of the United Nations (FAO) 2011b).

One of the beneficial characteristics of tilapia is the relative ease of genetic manipulation to improve their productivity. Hybridization is a relatively common practice with *Oreochromis niloticus* $\times$ *O. aureus* yielding mostly male populations that are capable of growing at faster rates than non-hybrid counterparts as well as controlling widespread reproduction within ponds. Similarly, hormone induced sex-reversal can also be performed. However, since initial worldwide 'founder populations' were both small and unknown, leading to inbreeding and poorer performance, genetic selection methods are underway to improve tilapia productivity with the development of the 'genetically

improved farmed tilapia' (GIFT) strain being an example of these efforts (reviewed by Eknath & Hulata 2009). Therefore, with the trend to increase tilapia productivity via genetic programmes coupled with tilapia aquaculture moving away from extensive to semi-intensive farming practices, there will inevitably be a greater reliance on designing species-specific (even strain-specific) and cost-effective diets to keep pace with their expansion. For example, in 1995, 70% of global tilapia production was based on the provision of commercial feeds, which increased to 85% in 2010 and it is anticipated that by 2020 this will further increase to 95% (Food & Agriculture Organization of the United Nations (FAO) 2011b). This underscores the importance of appropriate nutrition and feeding management, since the feeding cost is often a major expenditure in a modern tilapia farm (El-Sayed 1999; Ng *et al.* 2011). At present, the nutrient requirements and feeding of some of these improved tilapia strains are not fully understood and will be discussed in this review.

An obstacle to overcome with tilapia aquaculture is the rising cost and unpredictable reliability of fishmeal (FM) and fish oil (FO) global supplies. Currently, fishmeal and fish oil are included at between 0–20% and 0–10%, respectively, in the commercial diets of tilapia, depending on the country and culture system, with terrestrial-based protein, lipids and carbohydrates being used in higher proportions (Tacon & Metian 2008). Indeed, in 2010, the mean fishmeal use in commercial tilapia feeds was 3%, and is predicted to reduce to 2% and 1% in the years 2015 and 2020, respectively (Food & Agriculture Organization of the United Nations (FAO) 2011b). Focused research on identifying suitable alternatives, as either a partial or complete replacement, has been instrumental in reducing fishmeal

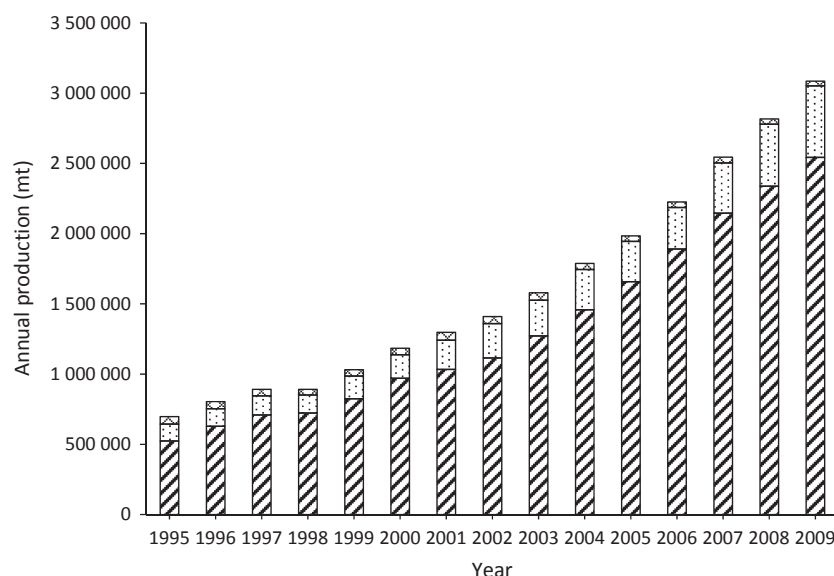


Figure 1 Worldwide annual production (metric tons, mt) of Nile tilapia (*Oreochromis niloticus*), Mossambique tilapia (*O. mossambicus*) and tilapia nei. Based on data from Food and Agriculture Organization of the United Nations (FAO) (2011a). ■, Mossambique tilapia; □, Tilapia nei; ▨, Nile tilapia.

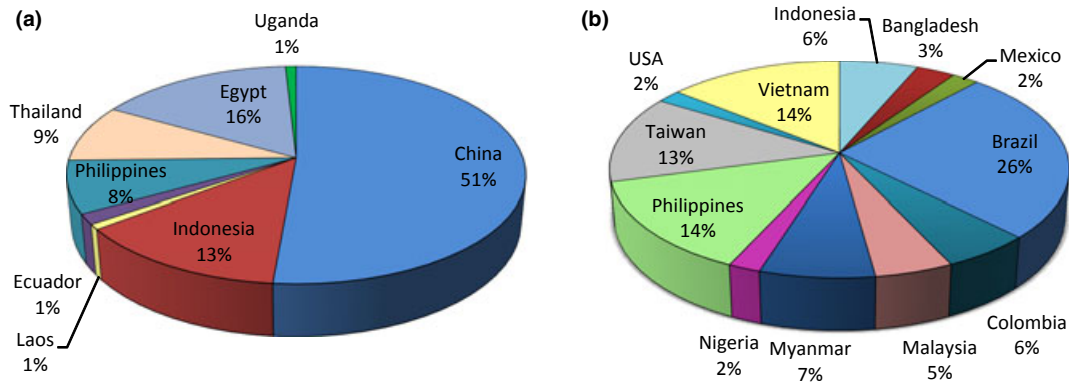


Figure 2 Top producers of Nile tilapia, *Oreochromis niloticus* (a) and other tilapia species not including *O. niloticus* (b). Based on data from Food & Agriculture Organization of the United Nations (FAO) 2011a.

use in tilapia feeds (e.g. El-Sayed 1999; El-Saidy & Gaber 2003, 2004; Furuya *et al.* 2004; Borgeson *et al.* 2006; Liti *et al.* 2006). For many fish species, including tilapia, fish-meal and fish oil often provide complete amino acid profile and fatty acid requirements, respectively. However, when investigating suitable alternatives, it becomes necessary fully to understand and quantify the specific requirements of tilapia as well as feeding management factors that may influence their nutrient utilization and thus productivity and fillet quality. Much published literature exists in this area; however, many of these novel ingredients evaluated are locally based, non-commercial and are of limited supply. For the purpose of this review, we will only focus on the few major alternative ingredients that are globally available for use in commercial tilapia feeds.

Several past reviews have focused on tilapia nutrition including identifying alternative protein sources (El-Sayed 1999), the lipid and fatty acid requirements (Lim *et al.* 2011b) and book chapter nutritional reviews (Jauncey 2000; Shiau 2002; El-Sayed 2006). These reviews were either published many years ago or focused on only one aspect of tilapia nutrition. The present review aimed to update the nutritional information, comprehensively focusing on the nutritional requirements of commercially important tilapia species. Since *O. niloticus*, *O. aureus*, *O. mossambicus* and their hybrids are the most farmed, these species will be the main focus of this review. However, when available, the nutritional requirements for other tilapia species will also be presented. Nutritional information and feeding management at various tilapia life stages from broodstock to finishing diets will be presented and discussed while commercial tilapia feeds and feeding management as practised at the farm level will also be highlighted. This single comprehensive review incorporating both research-based and practical information on tilapia nutrition and feeding management will likely benefit researchers, feed manufacturers, tilapia farmers, seafood

processors and policy makers involved in this rapidly expanding billion dollar industry.

Protein and amino acid requirements of tilapia

Proteins are composed of amino acids (AA) and generally represent the most expensive ingredient in fish diets (El-Sayed 1999). The total protein requirement can be influenced by the species, digestibility and amino acid profile of the protein source and life stage status. Generally, tilapia fry and spawning females have a higher protein requirement (30–40%) than the fingerlings and the grow-out stages (20–30%) for optimal performance (Siddiqui *et al.* 1998; Sweilum *et al.* 2005; Abdel-Tawwab *et al.* 2010). Water salinity between 0 and 10‰ had no significant effect on the dry matter, protein or energy digestibility of diets fed to Nile tilapia (De Silva & Perera 1984), while salinities between 1.2‰ and 25‰ had no significant effect on their total protein requirement (Larumbe-Morán *et al.* 2010). However, ultimately, the quality and essential amino acid (EAA) profile of the protein source will greatly influence the total protein requirement.

In order to maximize the growth benefit of protein, thereby minimizing cost, the ideal protein concept can be used where dietary amino acid is provided at the same ratio as required by the animal (Furuya *et al.* 2004; Nguyen & Davis 2009). Moreover, this will also reduce nitrogenous waste associated with amino acid being underutilized or deaminated as an energy source (Abdel-Tawwab *et al.* 2010). Determining the essential amino acid may be accomplished via dose–response experiments, where an amino acid is added or removed and the survival, growth or condition of the animal is evaluated while examining the amino acid profile of the whole fish may also help to predict their amino acid requirements (Ngamsnae *et al.* 1999).

In one of such dose–response studies, Santiago and Lovell (1988) quantified the essential amino acid requirements for Nile tilapia (Table 1). The ten essential amino acid for

tilapia, which are the same for other fish, include lysine (Lys), methionine (Met), tryptophan (Trp), arginine (Arg), phenylalanine (Phe), histidine (His), isoleucine (Ile), leucine (Leu), threonine (Thr) and valine (Val). Tyrosine (Tyr) and cysteine (Cys) are referred to as 'semi-essential' since Phe and Met can be synthesized from these amino acids, respectively. In the case of Met, the addition of Cys has been shown to reduce the requirement of Met by up to 49% on an equimolar sulphur basis (Nguyen & Davis 2009) and therefore these are often referred to as the total sulphur amino acids in nutritional studies.

It is widely regarded that fishmeal is the best protein source for tilapia, and other fish, due to their complete amino acid profile, high digestibility and residual lipid content that contains beneficial fatty acids. However, as previously mentioned, alternative protein sources are increasingly being used due to the rising cost and decreasing availability of fishmeal. Often these alternative protein sources are deficient in one or more limiting amino acid and therefore supplementation of synthetic (crystalline) amino acid becomes necessary to achieve optimal productivity. However, there have been conflicting reports on the efficiency of synthetic amino acid utilization for tilapia. It has been reported that when synthetic amino acid comprised 30% of the diets, tilapia poorly utilized this as a protein source (Gaye-Siessegger *et al.* 2007). It should be noted, however, that in this study, comparisons between synthetic and natural (i.e. protein bound) amino acids were not performed. Furthermore, it is unlikely synthetic amino acids would be used in such high amounts in practical diets for tilapia. In contrast, increasing dietary amounts of crystalline Met supplementation, from 0 to 0.24%, to meet the requirements of Nile tilapia significantly improved weight gain, indicating synthetic Met was utilized effec-

tively (Nguyen & Davis 2009). Depending on the amino acid profile of the protein source, the addition of appropriate amounts and the type of synthetic amino acid is often practised in commercial tilapia feed formulations.

To date, the vast majority of research on alternative protein sources has focused on terrestrial plants, and over 30 different species have been proposed as partial replacements to fishmeal. Since it is not feasible to discuss each one, only those that are generally accepted as suitable partial/complete replacements to fishmeal on a commercial scale will be discussed in the following sections. A separate section is devoted to the use of alternative protein mixtures. Table 2 summarizes the recommended dietary inclusion levels of protein alternatives that have relatively high success rates when evaluated in tilapia feeds.

Animal protein

The most commonly used animal protein alternatives to fishmeal include feather meal, meat bone meal (MBM) and animal/fishery by-products. Comparisons among otherwise discarded by-catches including mixed benthic meal, small pelagic meal, catfish meal, processing waste meal and commercially available anchovy meal showed that catfish meal provided significantly better growth to tilapia (Goddard *et al.* 2008). It was suggested that this was linked with the highest dietary amino acid levels, with the exception of Met, despite also having the highest ash content (Goddard *et al.* 2008). Such by-catches may have variable quality and ash content depending on the location, processing or composition and therefore results may not be universal.

Much of the research on animal proteins has focused on rendered terrestrial sources that have been used successfully in livestock and are thus commercially available (Bishop *et al.* 1995; Fasakin *et al.* 2005; Guimarães *et al.* 2008; Hernández *et al.* 2010). Bishop *et al.* (1995) reported that 66% of feather meal can replace a mixture of fishmeal and meat bone meal in the diets of Nile tilapia fry, with Ile being the only tested essential amino acid to be deficient, although a complete replacement was not feasible due to lower protein digestibility (Bishop *et al.* 1995). This was later confirmed in a separate study, when among fishmeal, feather meal, meat bone meal and poultry by-products, the protein digestibility of feather meal and meat bone meal was significantly lower to Nile tilapia compared with fishmeal and poultry by-products (Guimarães *et al.* 2008). Further, despite feather meal being hydrolysed or pretreated with enzymes, this actually resulted in significantly lower growth to hybrid tilapia when used as a partial replacement (66%) to high quality fishmeal, likely the result of a low His and Lys content as well as the low protein digestibility (Fasakin *et al.* 2005). Nevertheless, feather meal is reportedly high in the essential sulphur containing amino acids (i.e. Met and

Table 1 The essential and semi-essential amino acid (% of protein) for Nile tilapia, *Oreochromis niloticus*. Table modified from Santiago and Lovell (1988)

	% of protein
Essential amino acid	
Lysine	5.12
Methionine	2.68
Threonine	3.75
Tryptophan	1.00
Arginine	4.20
Phenylalanine	3.75
Histidine	1.72
Isoleucine	3.11
Leucine	3.39
Valine	2.80
Semi-essential amino acid	
Cystine	0.53
Tyrosine	1.79

Table 2 Major alternative protein sources to fish meal in the diets of tilapia. Growth, physiological condition and, when applicable, cost benefits were used as the criteria for the provided recommended levels

Protein source + (supplement)	Species and weight (g)	Tested levels replacing FM	Recommended levels	Duration and feeding frequency (times/day)	Reference
Animal sources					
Animal by-product	<i>O. niloticus</i> (0.1)	0–100%	75–100%	7 weeks (3)	Rodríguez-Serna <i>et al.</i> (1996)
Poultry by-product + phosphorus (P)	<i>O. niloticus</i> (9.5)	100%	100%	8 weeks (3)	Hernández <i>et al.</i> (2010)
Porcine meal + (P)	<i>O. niloticus</i> (9.5)	100%	100%	8 weeks (3)	Hernández <i>et al.</i> (2010)
Feather meal + (P)	<i>O. niloticus</i> (0.01)	0–100% (replace FM & MBM)	66%	42 days (2)	Bishop <i>et al.</i> (1995)
Plant sources					
Soybean meal (SBM)	<i>O. mossambicus</i> (30–50)	0–100%	75%	7 weeks (3)	Jackson <i>et al.</i> (1982)
SBM + (Met, Lys & P)	<i>O. niloticus</i> × <i>O. aureus</i> (100 to 300)	0–100%	100%	55–65 days (NR)	Viola <i>et al.</i> (1988)
SBM and FFSB + (Met & Lys)	<i>O. niloticus</i> (32.3)	100%	100%	17 weeks (3)	Goda <i>et al.</i> (2007)
SBM + (Met, Lys, Thr & P)	<i>O. niloticus</i> (5.3)	100%	100%	85 days (6)	Furuya <i>et al.</i> (2004)
SPC + Met	<i>O. niloticus</i> (<2)	0–100%	100%	8 weeks (6)	Zhao <i>et al.</i> (2010)
Okara meal + (P)	<i>O. niloticus</i> (2.6)	0–100%	75%	12 weeks (2)	El-Saidy (2011)
Cottonseed meal (CSM)	<i>O. mossambicus</i> (30–50)	0–100%	25–50%	9 weeks (3)	Jackson <i>et al.</i> (1982)
CSM + (Met & Lys)	<i>Oreochromis</i> sp. (11.8)	0–100%	<50%	16 weeks (3)	Mbahinzireki <i>et al.</i> (2001)
CSM (CC) + (Lys)	<i>O. niloticus</i> (20)	80%	80%	130 days (2)	El-Sayed (1990)
CSM (DC) + (Lys)	<i>O. niloticus</i> (20)	65%	65%	130 days (2)	El-Sayed (1990)
CSM + (Met & Lys) & 0–1858 mg/kg Fe	<i>O. niloticus</i> (3.7)	67%	67% + 886 to 1858 mg/kg Fe	30 weeks (NR)	El-Saidy and Gaber (2004)
Palm kernel meal (PKM)	<i>O. mossambicus</i> (8.4)	0–50% (replace SBM)	50%	12 weeks (2)	Lim <i>et al.</i> (2001)
Raw PKM	<i>Oreochromis</i> sp. (5.1)	0–40%	20%	10 weeks (2)	Ng <i>et al.</i> (2002)
Enzyme treated PKM	<i>Oreochromis</i> sp. (5.1)	0–40%	40%	10 weeks (2)	Ng <i>et al.</i> (2002)
Sunflower seed meal	<i>Tilapia rendalii</i> (9.3)	0–50%	20%	70 days (3)	Olvera-Novoa <i>et al.</i> (2002)
Sesame meal + (Met, Lys & P)	<i>O. niloticus</i> (8.7)	0–48% (replace FFSB)	8 & 16%	8 weeks (4)	Guo <i>et al.</i> (2011)
Rapeseed meal	<i>O. mossambicus</i> (0.3)	0–60% (replace FM and SBM)	15%	8 weeks (3)	Davies <i>et al.</i> (1990)
Canola meal	<i>O. niloticus</i> × <i>O. aureus</i> (6.3)	0–100% (replacing SBM)	19%	8 weeks (1)	Zhou and Yue (2010)
Ground nut cake	<i>O. niloticus</i> (4.2)	0–50%	50%	56 days (2)	Agbo <i>et al.</i> (2011)
Cowpea	<i>O. niloticus</i> (2.6)	0–100%	20–33%	70 days (2)	Keembiyehetty and De (1993)
Black gram seeds	<i>O. niloticus</i> (2.6)	0–100%	20–33%	70 days (2)	
Untreated cowpea	<i>O. niloticus</i> (0.2)	45%	45%	42 days (3)	Olivera-Castillo <i>et al.</i> (2011)
DHT cowpea	<i>O. niloticus</i> (0.2)	45%	45%	42 days (3)	Olivera
WHT cowpea	<i>O. niloticus</i> (0.2)	45%	45%	42 days (3)	Olivera
Pea protein isolate	<i>O. niloticus</i> (2.2)	30–60%	30%	56 days (NS)	Schulz <i>et al.</i> (2007)
Heat treated JPKM + (Lys)	<i>O. niloticus</i> (12–15)	62.5%	62.5%	12 weeks (5)	Akinleye <i>et al.</i> (2012)
Corn co-products + (Lys, Met, Thr & P)	<i>O. niloticus</i> × <i>O. aureus</i> (21)	0–100%	55%	10 weeks (NS)	Twibell and Brown (1998)
PPM + 0.5% Met and Lys	<i>O. niloticus</i> (3.7)	0–100%	75–100%	16 weeks (2)	El-Saidy and Gaber (2003)

Table 2 (continued)

Protein source + (supplement)	Species and weight (g)	Tested levels replacing FM	Recommended levels	Duration and feeding frequency (times/day)	Reference
'Simple' PPM	<i>O. niloticus</i> (12)	0–100%	<100%	64 days (2)	Borgeson <i>et al.</i> (2006)
'Complex' PPM	<i>O. niloticus</i> (12)	0–100%	100%	64 days (2)	Borgeson <i>et al.</i> (2006)
Guar seed meal	<i>O. niloticus</i> (6.9)	0–75%	50%	84 days (2)	Al-Hafedh and Siddiqui (1998)

NR, not reported; FFSB, full fat soybean meal; SPC, soybean protein concentrate; CC, corticated cottonseed; DC, decorticated cottonseed; Dehulled heat treated (48°C for 7 h, 70°C for 5 h and 119°C for 30 min); WHT, whole seeds heat treated (48°C for 7 h, 70°C for 5 h and 119°C for 30 min); JPKM, *Jatropha platyphylla* kernel meal (defatted kernel meal heat treated 121°C for 20 min.); PPM, consisted of SBM, CSM, sunflower meal, linseed meal all in equal proportions of 25%. PPM, plant protein mixture. Simple plant protein mixture (PPM) consisted of wheat, maize gluten corn and SBM and the complex plant protein mixture (PPM) consisted of wheat, maize gluten corn, SBM, canola protein concentrate, dehulled flax and pea protein concentrate.

Cys) and may be beneficial when used in a mixture with other protein sources (Fowler 1990; Fasakin *et al.* 2005) and this is discussed below.

El-Sayed (1998) found that among various animal sources, that included bloodmeal, meat bone meal and poultry by-products, the best growth and feeding efficiency was obtained with bloodmeal, followed by poultry by-product and both led to a better profit index than fishmeal. Meanwhile, a total fishmeal replacement with commercially available poultry by-product meal-petfood grade-based diet (PBM-PG) was possible, although in the same study, porcine meal (PM) resulted in a significantly lower growth and feeding efficiency (Hernández *et al.* 2010). Hernández *et al.* (2010) suggested that this was linked with the high apparent digestibility coefficient (ADC) of crude protein and energy from fishmeal (89.7% and 86.0%, respectively) and PBM-PG (87.8% and 84.4%, respectively) while those of porcine meal were lower (82.1% and 79.6%, respectively).

Plant proteins

The most common plant protein used in tilapia diets is soybean meal (SBM), however, due to the rising cost of this ingredient, alternatives are also being investigated that include cottonseed meal (CSM) and rapeseed/canola meal (R/CM), and the level of inclusion will largely depend on the country where tilapia is farmed. Figure 3 shows the production of soybeans, cottonseed and rapeseed from the highest producing countries in 2009 (Food & Agriculture Organization of the United Nations (FAO) 2011c).

Soybean meal and soybean protein concentrates

Soybean meal can be included up to 60% depending on the country (Food & Agriculture Organization of the United Nations (FAO) 2011b) and its popularity is due to its lower cost, high protein content (40–44%) and relatively good amino acid profile (El-Sayed 1999) where many studies

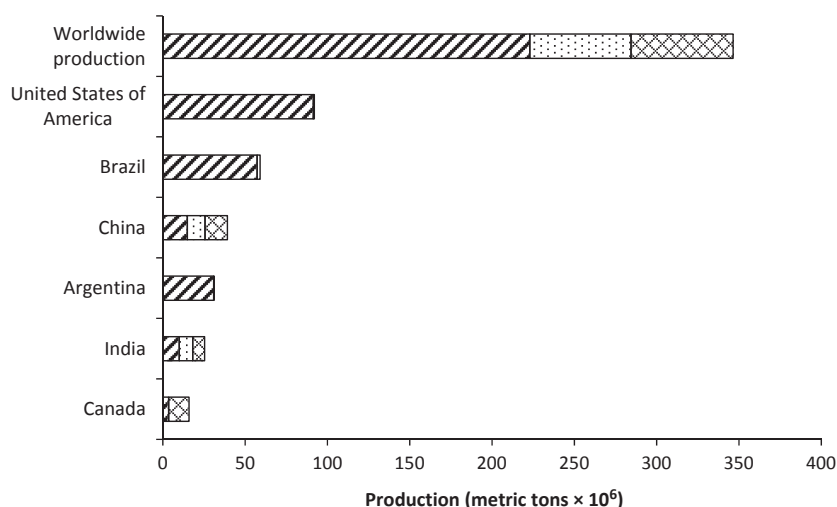


Figure 3 The annual production of soybeans, cottonseed and rapeseed from the top producing countries in 2009. Data based on data from Food and Agriculture Organization of the United Nations (FAO) (2011c). ▨ Soybeans; ▤ Cottonseed; ▩ Rapeseed

have also shown a high digestibility in tilapia (Sintayehu *et al.* 1996; Riche *et al.* 2001; Guimarães *et al.* 2008; Zhou & Yue 2012). For example, with tilapia hybrids, the apparent digestibility (AD) of protein from fermented soybean meal and soybean meal (94.9% and 90.9%, respectively) was often significantly higher than diets containing canola meal, cottonseed meal, peanut meal or various animal sources including fishmeal (Zhou & Yue 2012) while Riche *et al.* (2001) detected no significant differences in the apparent amino acid availability (AAAA) between graded levels of dietary soybean meal (without added phytase) and dietary fishmeal. Meanwhile, Guimarães *et al.* (2008) showed that the AAAA from soybean meal was either not significantly different or significantly higher than those of corn gluten meal, CSM28 (28% protein) or CSM38 (38% protein), while the apparent digestibility of protein from soybean meal (92.4%) was actually higher than fishmeal (88.6%) for Nile tilapia.

While the protein and amino acid digestibility from soybean meal is high, it is generally deficient in the sulphur essential amino acid (Met and Cys) that are required by tilapia for optimal growth (Furuya *et al.* 2004). In an early study with soybean meal, no amino acid supplementations were performed and a low dietary Met content was implicated as one of the main causes for depressing the growth in Mozambique tilapia (Jackson *et al.* 1982). Subsequently, L-lysine and DL-methionine supplementations have been performed often to ensure that they do not become limiting (Riche *et al.* 2001; Furuya *et al.* 2004; Goda *et al.* 2007), although sometimes this is still not enough for soybean meal to completely replace fishmeal (Viola *et al.* 1988; Sintayehu *et al.* 1996; Furuya *et al.* 2004; Goda *et al.* 2007).

In a series of feeding experiments within ponds (duration 30–54 days) the growth and feed conversion ratios (FCR) of tilapia hybrid (*O. niloticus* × *O. aureus*) were analysed when fed soybean meal-based diets using different amino acid and/or phosphorus (P) supplementations, and Viola *et al.* (1988) concluded that P was the only necessary supplement to achieve comparable results to those fed fishmeal-based diets. However, since this experiment was conducted in ponds, natural food present may have provided additional nutrition. In the absence of P supplementation, Met and Lys additions did not improve tilapia growth or feeding efficiency when fed soybean meal-based diets (Sintayehu *et al.* 1996). In another study, soybean meal or full-fat soybean-based diets with amino acid supplementation were fed to Nile tilapia and Galilae tilapia, *Sarotherodon galilaeus*, and the absence of P addition was suggested to be the cause for the lower growth of Nile tilapia, although the growth of Galilae tilapia was unaffected (Goda *et al.* 2007). Using a combination of amino acid and P supplementations to soybean meal-based diets, Furuya

et al. (2004) concluded that both amino acid and P supplementations were necessary completely to replace fishmeal to ensure optimal survival, growth feeding efficiency and fillet yield. Due to the importance of P to tilapia, dicalcium phosphate (DCP) is often added (2.9–15 g kg⁻¹ dry diet) to SBM-based diets to ensure this does not become a limiting nutrient (Furuya *et al.* 2004; Nguyen & Davis 2009).

One of the reasons for mineral and amino acid supplementation when soybean meal is used is because of the presence of phytic acid, an antinutritional factor (ANF) associated with some plant sources including soybeans. Phytic acid is used principally to store P in plants, while phytate in an anionic salt form and has the ability to bind with divalent cations (e.g. Ca²⁺, Fe²⁺, Mg²⁺). In addition to directly reducing mineral availability to fish, phytate can directly reduce protein availability by forming binary and tertiary complexes with proteins and minerals, as well as causing intestinal damage to fish (Francis *et al.* 2001). While heat can potentially reduce phytic acid (Francis *et al.* 2001), the exogenous application of dietary phytase, an enzyme responsible for hydrolysing phytate, has been proposed since phytase was suggested to be lacking in the intestine of tilapia (Riche *et al.* 2001; Riche & Garling 2004).

However, there have been conflicting reports on the efficacy of dietary phytase supplementation (Riche *et al.* 2001; Lavorgna *et al.* 2003; Riche & Garling 2004; Liebert & Portz 2005). Lysine and alanine (Ala) digestibility significantly decreased for Nile tilapia fed phytase supplemented SBM-based diets, compared with those fed diets with untreated soybean meal (Riche *et al.* 2001). Riche and Garling (2004) also revealed that the presence of phytic acid had no significant effect on nitrogen retention in Nile tilapia and, moreover, phytase treated soybean meal reduced the apparent protein utilization. It was subsequently demonstrated that phytase activity did exist within tilapia intestines, although phytate concentrations exceeding 3.5 mM had an inhibiting effect *in vitro* (Lavorgna *et al.* 2003). Furthermore, both inorganic and organic P, originating from defluorinated P and inorganic P from dietary soybean meal, wheat middlings and corn grain had a high digestibility of 93.2% and 90.0%, respectively, leading to the suggestion that dietary phytase supplementation is not necessary in tilapia feeds (Lavorgna *et al.* 2003). In contrast, Liebert and Portz (2005) found that dietary phytase activities of 1250 FTU kg⁻¹ caused a significantly higher growth performance and nutrient utilization (protein, energy and P) to Nile tilapia compared with those fed diets with and without P supplementation and with no phytase additions. The form of phytase used in all these studies were the same and further research is required to reconcile these contradictory findings. Recently, it was shown that fermented soybean meal had significantly

higher P and Met availability to tilapia than soybean meal (Zhou & Yue 2012) although the potential effects on phy-tate or other antinutritional factors during the fermenting process are unclear.

Another antinutritional factors associated with soybean meal is saponin (Francis *et al.* 2001). While the toxicity and digestive effects of saponin to fish is well documented, studies on soya-saponin (e.g. galacto-oligosaccharides, raffinose, stachyose and verbascose) that exist in different concentrations and forms, have received relatively little attention in tilapia nutrition and have mainly focused on other fish species (Chen *et al.* 2011; Sørensen *et al.* 2011). Therefore, due to the popularity of soybean meal, experiments using different soya-saponin types, concentrations and potential interactions with other antinutritional factors are warranted to determine their potential effects in tilapia.

The use of soybean protein concentrates (SPC) has been proposed as an alternative to soybean meal due to their higher crude protein and lower ANF content, although it can be higher in fibre (Lusas & Riaz 1995; Zhao *et al.* 2010). Zhao *et al.* (2010) examined the feasibility of replacing fishmeal with soybean protein concentrates by feeding Nile tilapia (GIFT strain) fry (<2 g) twice per day with soybean protein concentrates supplemented with Met (SPCM2) or fishmeal-based diets and six times per day with and without Met supplementation (SPC6 and SPCM6, respectively) (Zhao *et al.* 2010). Despite having similar Met levels, tilapia in the SPCM2 treatment group had a significantly lower final body weight, specific growth rate (SGR), feed intake, FCR, protein efficiency ratios (PER) and productive protein value (PPV) than tilapia fed fishmeal-based diets (Zhao *et al.* 2010). It was concluded that for soybean protein concentrates-based diets to totally replace fishmeal, feeding frequencies needed to be increased from twice to six times per day along with Met supplementation, leading to the suggestion that a lower Met utilization efficiency of synthetic Met or a higher feeding frequency inducing an elevated digestion of carbohydrates and lipids may have been the cause (Zhao *et al.* 2010). The reported ability of tilapia fry to utilize soybean protein concentrates after dietary and feeding modification is important considering that the fry stage is still highly dependent on fishmeal-fish oil-based diets to ensure their optimal growth during this relatively short but sensitive culture period. The feeding management of the tilapia fry stage will be discussed in more detail later.

Cottonseed meal

Cottonseed meal has been investigated as a potential alternative to both fishmeal and soybean meal due to its cheaper cost, being readily available in some countries, particularly in China, India and Egypt, although the protein content

can be variable (23–53%) depending on how this product is processed (El-Sayed 1999; El-Saidy & Saad 2011). It has been reported that China currently uses cottonseed meal in commercial tilapia diets at between 0 and 25% (Food & Agriculture Organization of the United Nations (FAO) 2011b). Jackson *et al.* (1982) showed that increasing dietary cottonseed meal (commercial pre-pressed solvent extracted meal), caused no significant growth difference to Mozambique tilapia compared with other plant-based meals including soybean meal, rapeseed meal and sunflower meal. Despite no amino acid or P supplementations, the reasons for the relatively good utilization of cottonseed meal by tilapia was suggested to be the low free gossypol levels (0.02%) leading to Lys being readily available at 92.2% (Jackson *et al.* 1982). Indeed, several researchers have stated that the success of utilizing dietary cottonseed meal in tilapia feeds will ultimately depend on the gossypol levels and available Lys, since the former can bind with the latter, while gossypol can also produce intestinal abnormalities in various fish species (Francis *et al.* 2001; El-Saidy & Gaber 2004; Garcia-Abiado *et al.* 2004). Processing could potentially minimize gossypol levels, with adding acetone to the extract oil, reportedly having a better capacity to remove gossypols than other solvents (Cheng & Hardy 2002).

Mbahinzireki *et al.* (2001) found that cottonseed meal inclusion levels of 75% (total and free gossypols of 7.38 and 0.70 g kg⁻¹ diet, respectively) and 100% (total and free gossypols of 9.16 and 0.99 g kg⁻¹ diet, respectively) significantly reduced the growth, feed conversion ratio and P content of *Oreochromis* sp. Garcia-Abiado *et al.* (2004) investigated the underlying cause of gossypols affecting *Oreochromis* sp. growth and found this was linked to gossypols lowering the haematocrit and haemoglobin levels as well as causing spleen abnormalities. Subsequently, a dietary cottonseed meal level of 29.4% was recommended for tilapia (Garcia-Abiado *et al.* 2004) while Yue and Zhou (2008) similarly reported that dietary cottonseed meal exceeding 33.7% tended negatively to impact growth, feeding efficiencies and haematological values.

A different interpretation for reduced growth when using cottonseed meal was offered in experiments by El-Sayed (1990) and El-Saidy and Saad (2011). When comparing corticated cottonseed meal (CC) and decorticated cottonseed meal (DC), it was shown that although DC resulted in significantly higher growth and feeding efficiency than CC, these were both inferior to tilapia fed fishmeal-based diets (El-Sayed 1990). Although dietary gossypols were not measured, El-Sayed (1990) suggested that gossypols or Lys availability could not explain the reduced growth rates, but rather the high fibre content in cottonseed meal. Nevertheless the profit index, taking into consideration fillet yield and feeding costs, was significantly higher for those fed CC, followed by DC and it was suggested that 50% of

cottonseed meal could replace fishmeal (El-Sayed 1990). Moreover, despite Lys supplementations, a 100% replacement of dietary soybean meal by cottonseed meal (solvent-extracted) in Nile tilapia diets significantly reduced their growth, feed intake and feeding efficiency compared with fish fed diets with 0 (with soybean meal as the sole protein source), 25%, 50% or 75% cottonseed meal (El-Saidy & Saad 2011). Since the maximum dietary free gossypols levels (45.69 mg kg^{-1} dry matter) were far below levels reported to reduce growth in blue tilapia, *O. aureus* (180 mg kg^{-1} dry matter) (Robinson *et al.* 1984), El-Saidy and Saad (2011) suggested that the reduced apparent digestibility of dry matter, protein and P were the likely reasons. It is interesting to note that de-gossypoled cottonseed meal had a higher apparent digestibility of dry matter, crude protein, lipid, energy and significantly higher P than cottonseed meal to hybrid tilapia (Zhou & Yue 2012), however, the processing method or gossypol concentrations were not mentioned.

While Lys supplementation is typical in many of these studies, dietary ferrous sulphate (FeSO_4) can have a neutralizing effect on gossypols (El-Saidy & Gaber 2004). It was shown that dietary Fe: gossypol ratios of 1:1, 1.5:1 and 2:1, resulted in no significant growth effect to Nile tilapia when cottonseed meal completely replaced fishmeal after 30 weeks (El-Saidy & Gaber 2004). Furthermore, ratios of 1.5:1 and 2:1 significantly improved ADC of protein, fat, dry matter and energy as well as blood parameters (El-Saidy & Gaber 2004). Based on investigations with terrestrial animals, it is believed that Fe binds with gossypols thus minimizing their intestinal absorption (Wedegaertner 1981). Since gossypol levels may vary drastically in cottonseed meal based-diets, depending on cottonseed meal inclusion and gossypol levels of cottonseed meal itself (Jackson *et al.* 1982; Mbahinzireki *et al.* 2001; El-Saidy & Gaber 2004), this ratio provides a useful value when formulating practical tilapia diets. However, it must be noted that since high dietary Fe can cause adverse immunological responses (Sealey *et al.* 1997), an optimal balance must be met and warrants further research.

Rapeseed/canola meal

Rapeseed/canola meal (R/CM) is a commonly used protein alternative in the diets of tilapia in China (between 20 and 40%) (Food & Agriculture Organization of the United Nations (FAO) 2011b). Rapeseed/canola meal has a relatively high protein content as well as being generally cheaper than fishmeal or SBM, however, there are substantially fewer investigations compared with either soybean meal or cottonseed meal (Jackson *et al.* 1982; Davies *et al.* 1990; Zhou & Yue 2010; Yigit & Olmez 2011). Jackson *et al.* (1982) found that rapeseed meal provided similar growth rates and feeding efficiency in Mozambique tilapia

to soybean meal and cottonseed meal when used between 25% and 50%. In contrast, Davies *et al.* (1990) reported that beyond an inclusion of 15% (substituting for both fishmeal and soybean meal) led to lower growth and feeding efficiencies in Mozambique tilapia and this was suggested to be linked with relatively high glucosinolate levels, despite being pretreated at 90°C as well as mechanically treated and solvent extracted (Davies *et al.* 1990). Since the glucosinolate levels in the studies by Jackson *et al.* (1982) and Davies *et al.* (1990) were similar (7.5 and 6.2 mg g^{-1} , respectively), this may indicate that other antinutritional factors were also responsible for reducing tilapia growth.

Zhou and Yue (2010) suggested that glucosinolates, phytate, tannins and/or fibre in canola meal were responsible for reducing hybrid tilapia growth and feeding efficiencies when used at levels exceeding 45%. To potentially mitigate the higher fibre content, exogenous cellulase additions were made ($1\text{--}5 \text{ g kg}^{-1}$) to Nile tilapia diets, although the high canola meal diet, with or without cellulose, still significantly reduced their growth and nutrient digestibility (Yigit & Olmez 2011). Research should continue to examine the extent of antinutritional factors influencing productivity and the various potential mitigating methods when R/CM is used in tilapia feeds.

Mixtures of alternative proteins

Generally, the majority of studies that recommended only partial replacements of alternative protein sources to fishmeal implicate antinutritional factors, an incomplete amino acid profile, P deficiencies or reduced digestibility/utilization as the reasons (El-Sayed *et al.* 2000; El-Saidy & Gaber 2004; Garcia-Abiado *et al.* 2004; Yue & Zhou 2008; Zhao *et al.* 2010). Rather than continuously investigating one protein source, several studies have examined the feasibility of utilizing different animal and/or plant mixtures in tilapia feeds. While the dietary ingredient combinations will vary, the ultimate goal here is to dilute potential antinutritional factors associated with one protein source as well as complementing the overall amino acid profile (El-Saidy & Gaber 2003; Borgeson *et al.* 2006; Liti *et al.* 2006).

A combination of dried distiller grain with solubles (DDGS), meat bone meal and soybean meal was demonstrated to be able fully to replace a combination of fishmeal and soybean meal in the diets of tilapia hybrids without reducing productivity, despite both Met and Lys being deficient (Coyle *et al.* 2004). Similarly, blood meal, meat bone meal, feather meal and fishmeal mixtures could successfully replace the sole use of high quality fishmeal, and although growth performance and feeding efficiency was slightly lower, it was not significantly different (Rodríguez-Serna *et al.* 1996). This is particularly noteworthy considering no

amino acid supplementations were made despite Lys being markedly lower compared with the fishmeal only diets (Rodríguez-Serna *et al.* 1996).

The feasibility of relying completely on plant protein mixtures (PPM) has also been investigated with good success. Increasing dietary PPM levels of 25%, 50%, 75% or 100%, comprising soybean meal, cottonseed meal, sunflower meal and linseed meal (all in equal proportions of 25%) and supplemented with 0.5% Met and Lys, were fed to Nile tilapia fingerlings and compared with those fed a 100% fishmeal diet over 16 weeks (El-Saidy & Gaber 2003). It was reported that no significant differences in tilapia growth performance or feeding efficiencies were detected among the various diets, despite nutrient digestibility being lower in diets with higher plant protein mixtures levels (El-Saidy & Gaber 2003). Moreover, the cost-benefit analysis increased with increasing dietary plant protein mixtures use with the 100% plant protein mixture diet being over 1.7 more cost effective than fishmeal-based diets (El-Saidy & Gaber 2003). In another plant protein mixture study, Borgeson *et al.* (2006) investigated the effects of 'simple' and 'complex' plant protein mixtures on the growth and feeding efficiencies of Nile tilapia against those fed a fishmeal-based diet for 63 days. The simple mixture diets consisted of maize gluten meal and soybean meal at 33%, 67% or 100%, while the three complex mixtures consisted of maize gluten meal, soybean meal, canola protein concentrate, dehulled flax and pea protein concentrate at 33%, 67% or 100%. While each diet had different plant protein proportions, these were all designed to have similar digestible protein and energy contents, and no amino acid or P supplementations were performed. It was demonstrated that tilapia fed the complex plant protein mixture diet had significantly better growth, PER and feed/gain ratios than those fed the simple plant protein mixture diet while the growth performance and nutrient utilization efficiencies were not significantly different from tilapia fed the fishmeal-based diets (Borgeson *et al.* 2006). Borgeson *et al.* (2006) suggested that the success of utilizing a 100% complex plant protein mixture diet was due to the relatively high dietary protein coupled with increased gross energy and dry matter digestibility due to reduced ANF. A number of factors presumably led to this, including the utilization of protein concentrates that have lower ANF, the diluting effect of any ANF present as well as both the simple and complex PPM-based diets containing an amino acid content in excess of the requirement for Nile tilapia (Borgeson *et al.* 2006). A similar diluting effect of plant protein ANF was also suggested for good tilapia growth when using a mixture of dietary cottonseed meal, wheat bran and shrimp head meal over 250 days in a study by Liti *et al.* (2006). However, since Nile tilapia were cultured within fertilized ponds, it was suggested that supplemental nutrition from

natural phytoplankton present in the ponds may have a contributing factor.

While the majority of plant proteins are only capable of partially replacing fishmeal (or soybean meal), these dietary mixture studies show the potential of utilizing ingredients without the need for extra dietary supplementation. Moreover, through deliberate considerations to the amino acid and mineral content, it may be possible to take advantage of more locally available and cheap protein sources (see Table 2) further to reduce feeding costs. Research is certainly warranted to continue exploring this area, such as quantifying the ANF composition, amino acid profile and protein digestibility/utilization of such ingredient mixtures to offer more flexibility to tilapia farmers and feed manufacturers rather than relying on one or two major protein sources.

Lipid and fatty acid requirements of tilapia

Lipids comprise five classes that include waxes, sterols, triacylglycerides, sphingolipids and phosphoglycerides (or phospholipids) and are classified as either polar or non-polar lipids depending on water solubility (National Research Council (NRC) 2011). In fish diets, lipids are added as a highly digestible and concentrated energy source as well as being important components to cellular membranes, precursors to hormones and aids in the absorption of lipid-soluble vitamins. Moreover, diet quality is also improved where the oil coating to extruded pellets enhances the palatability and appearance to fish (National Research Council (NRC) 2011). One technique to ensure energy does not become limiting, and consequently a catabolism of protein as an energy source, is via protein sparing whereby dietary non-protein energy is increased. Using gross energy calculations (MJ kg^{-1}) from Brett (1973), the optimal protein to energy (P/E) ratio, when using lipid as the energy source, has been investigated showing 19.7 mg crude protein (CP) kJ g^{-1} for Mozambique tilapia (El-Dahhar & Lovell 1995), 15.3 mg CP kJ g^{-1} and 16.4 mg CP kJ g^{-1} for 22.9 g and 39.8 g Nile tilapia, respectively (Sweilum *et al.* 2005) and 27.3 mg CP kJ g^{-1} for hybrid tilapia (*O. niloticus* $\times$ *O. aureus*) (Gao *et al.* 2011).

Appropriate dietary lipid inclusion levels are important considerations since deficiencies or excesses can reduce growth or produce an excessively fatty fish, respectively. The minimum requirement and optimal level has been shown to be 5% and 12% dietary lipid, respectively, in the feeds for hybrid tilapia (De Silva *et al.* 1991; Chou & Shiau 1996). Moreover, at various tested protein levels of 6–30%, increasing dietary lipid inclusions from 6% to 18% linearly increased protein sparing in all diets, while a lipid level of 24% decreased protein sparing (De Silva *et al.* 1991). On the other hand, if sufficient energy sources of digestible carbohydrates and proteins are present in the diets of

Mozambique tilapia, lower lipid levels could be used during intensive culture in recirculating systems (Fitzsimmons *et al.* 1997). Similarly, El-Sayed and Garling (1988) found that carbohydrates can have a sparing effect on lipids, however, detrimental growth may occur when excessive carbohydrates are present (carbohydrate to lipid ratio of 24.11), likely the result of a deficiency of dietary essential fatty acids. Therefore a balance must be met in which ideally carbohydrates are used as a cheaper energy source to maximize growth, while lipids are used to satisfy the essential fatty acid requirement of tilapia.

The major polyunsaturated fatty acids (PUFA) (<20 carbon chains) required by tilapia are α -linolenic (α LNA, 18:3n-3) and linoleic (LOA, 18:2n-6) acids since these cannot be synthesized by fish, or any vertebrate, and therefore these essential fatty acids must be provided in the diet. It has been reported that tilapia has the ability to synthesize the more physiologically important long chain polyunsaturated fatty acids (LC-PUFA) including eicosapentaenoic acid (EPA, 20:5n-3) and docosahexaenoic acid (DHA, 22:6n-3) from α -linolenic and arachidonic acid (ARA, 20:4n-6) from linoleic (Olsen *et al.* 1990; Tocher *et al.* 2002; Teoh *et al.* 2011). However, the elongation and desaturation of these essential C18 fatty acids to their LC-PUFA end products are likely insufficient to supply the required amounts and therefore dietary supplementation may be required. As such, studies have often focused on the n-3 and n-6 PUFA (and their ratios) as well as LC-PUFA requirements of tilapia. Even though fish oil often provides superior survival and growth to many fish species, some studies on tilapia have shown that high dietary concentrations of LC-PUFA (found in fish oil) may have detrimental effects on their growth. Although this may be related to the general trend of warm water fish having a higher requirement for n-6 PUFA than n-3 PUFA (National Research Council (NRC) 2011), the results for the fatty acid requirements of tilapia have, to date, been contradictory.

The fatty acid requirement of Nile tilapia was reported to be 0.5% for linoleic (Takeuchi *et al.* 1983) and for the red-belly tilapia, *Tilapia zillii*, to be approximately 1% for both linoleic and arachidonic acid (Kanazawa *et al.* 1980). It has been shown that high dietary n-3 PUFA reduced the growth of tilapia (Kanazawa *et al.* 1980; Huang *et al.* 1998) while >1% dietary α -linolenic lowered the growth of blue tilapia (Stickney & McGeachin 1983a). There was no growth improvement in tilapia when either dietary α -linolenic or n-3 PUFA were provided (Takeuchi *et al.* 1983) and the use of conjugated α -linolenic between 0.5–1.0% and 0.5–2.5%, at the expense of soybean oil and fish oil, respectively, had no significant effect on the growth performance or feed utilization of Nile tilapia or their GIFT strain (San tos *et al.* 2011; Luo *et al.* 2012). Furthermore, tilapia hybrid

fed diets with cod liver oil (CLO) (rich in n-3 LC-PUFA) had significantly lower feeding efficiency than those fed diets with crude palm oil (CPO), which is characterized by a high saturated fatty acid (SFA) and monounsaturated fatty acid (MUFA) content (Ng *et al.* 2001). In contrast, Chou *et al.* (2001) found that cod liver oil up to 5% had no growth depression in hybrid tilapia suggesting that n-3 PUFA was also important for maximal growth. These discrepancies in the n-3 PUFA requirement of tilapia may be due a wide variety of factors such as differences in age, nutritional history, experimental duration, water temperature, etc. The quantitative n-3 fatty acid requirement of tilapia is currently not known.

Traditionally, commercial tilapia feeds have depended mostly on marine fish oil as the sole dietary lipid source. As fish oil is becoming more expensive and less available (Turchini *et al.* 2009), it is crucial to identify suitable lipid alternatives in tilapia feeds. Currently, soybean oil (SBO) is commonly used as an alternative to fish oil since a number of studies have shown this to be viable for tilapia feeds (Stickney & McGeachin 1983b; Gaber 1996; Huang *et al.* 1998). An early study demonstrated the feasibility of soybean oil replacing fish oil since there were no significant growth differences in blue tilapia when the diets contained 10% menhaden oil, catfish oil or soybean oil (Stickney & McGeachin 1983b), while Gaber (1996) reported good growth for Nile tilapia when fed SBO-based diets. Furthermore, using semi-purified casein-based diets, Huang *et al.* (1998) reported that hybrid tilapia fingerlings fed 8% SBO or fish oil diets for 10 weeks showed no differences in growth or feed utilization. In contrast, after 6 weeks, Nile tilapia fed diets with 50% substitution of fish oil by soybean oil and linseed oil (LO) led to significantly lower growth than fish fed with fish oil-based diets (Szabó *et al.* 2011). Other than SBO, diets containing coconut oil, LO, grape-seed oil or poultry fat, at a 1:1 ratio with fish oil, did not significantly affect the growth of Nile tilapia over 15 weeks (Trushenski *et al.* 2009).

Since alternative lipid sources generally can be used successfully in tilapia feeds, there is also increasing interest in the use of palm oil since it is the least expensive of all the major vegetable oils and a readily available lipid source, being the most produced vegetable oil in the world (Ng *et al.* 2001, 2006; Bahurmiz & Ng 2007; Ng and Gibon 2011). The use of crude palm oil (CPO) and crude palm kernel oil (CPKO) in the diets of Nile tilapia did not significantly reduce their growth performance (Ng *et al.* 2001). Ng *et al.* (2006) found that palm-oil laden spent bleaching clay (SBC), which is a discarded by-product when palm oil is refined, could successfully replace 100% of added fish oil in the diets of Nile tilapia. Similarly, when raised from stocking to market size, Bahurmiz and Ng (2007) found that crude palm oil, palm fatty acid distillate or refined,

bleached and deodorized palm olein (RBDPO), which reduced the n-3/n-6 PUFA ratio from 2.8 (in fish oil-based diets) to between 0.5 and 0.6, did not significantly affect the growth performance, feeding efficiency or fillet/carcass yield of red hybrid tilapia. However, while the apparent digestibility of protein and dry matter were unaffected by the dietary lipid source, the apparent digestibility of lipid was lower in the diets containing palm oil-based diets due to SFA being less digestible than PUFA (Bahurmiz & Ng 2007). An added benefit of using crude palm oil in tilapia diets is a greater deposition of vitamin E in the tissues thereby enhancing the oxidative stability of the fillets (Wang *et al.* 2006).

With the increasing use of plant oils, the fatty acid composition of the final product for human consumption will always be of ubiquitous concern in terms of nutritional quality and is an obstacle to overcome (Huang *et al.* 1998; Bahurmiz & Ng 2007; Young 2009). 'Finishing diets' that can mitigate this problem as well as the influence of using the right tilapia strain (genetics) is discussed later.

Carbohydrates

Carbohydrates are categorized as monosaccharides, disaccharides, oligosaccharides or polysaccharides and unlike protein and lipids, fish have no specific requirement for these nutrients (Wilson 1994). Nevertheless, carbohydrates provide nutritive benefits including being an inexpensive alternative source of energy to protein (i.e. protein sparing) (Shiau & Peng 1993) thus improving growth (Anderson *et al.* 1984) as well as improving the stability of pelleted feeds. In practical terms for fish, carbohydrates can be broadly categorized as starches (simple and complex) and non-starch polysaccharides (NSP). Starches include dextrin and glucose which are relatively well digested, and in contrast, non-starch polysaccharides are utilized much more poorly or not at all (Francis *et al.* 2001; Sinha *et al.* 2011). However, the presence of non-starch polysaccharides are highly significant since their levels will inevitably increase with increasing use of plant-based feed ingredients, and may exert antinutritional properties to tilapia (Francis *et al.* 2001). This will be discussed later in this section.

In comparison with terrestrial animals, fish utilize carbohydrates less efficiently, although warm water herbivorous fish are better adapted than marine carnivores or cold water fish, likely reflecting the natural diets in their respective habitats and hence enzymatic activity (Wilson 1994). Indeed, in the past, fish have been referred to as being 'diabetic' due to an extended duration of hyperglycaemia when fed high starch diets, however, it is believed that rather than this being the result of inadequate insulin levels, it may be due to disrupted hormone-receptor activity (Wilson 1994; Shiau 1997). Thus it has been suggested that since glucose

is absorbed more quickly than the mobilization of enzymes responsible for their metabolism, this consequently leads to excessive blood glucose levels and therefore excretion in the urine of tilapia (Tung & Shiau 1993; Lin *et al.* 2000). For slower metabolizing complex sugars, such as starch, dextrin and disaccharides, it is well known that these are utilized more effectively in tilapia than glucose based on growth performance and feeding efficiencies (Anderson *et al.* 1984; Shiau & Suen 1992; Shiau & Peng 1993; Tung & Shiau 1993; Shiau & Chuang 1995) as well as contributing to increased energy storage (Hsieh & Shiau 2000). However, it must be noted that the exact mechanisms of carbohydrate utilization efficiencies are not yet completely understood in tilapia.

While the ability of tilapia to digest and utilize starches is largely dependent on the complexity of the starch, the presence of intestinal bacteria has also been proposed as a factor (Leenhouwers *et al.* 2007, 2008). Kihara and Sakata (1997) demonstrated that among various indigestible carbohydrate sources including α -starch, cellulose, sodium alginate, chitin and kaolin, Nile tilapia had intestinal bacteria capable of fermenting α -starch at higher rates than others, which produced short chain volatile fatty acids (VFA). Moreover, despite bacteria being capable of cellulolytic activity in the gut of Mozambique tilapia, it was pointed out that this would be unlikely to digest cellulose appreciably due to the insufficient duration before being excreted (Saha *et al.* 2006), and likely explains this carbohydrate as being non-fermentable (Kihara & Sakata 1997). Indeed, cellulose is regarded as having no nutritional benefit and is often added as an inert filler during nutrition experiments (Anderson *et al.* 1984; Sinha *et al.* 2011). Leenhouwers *et al.* (2008) found that among glucose, wheat starch, arabinoxylan and whole wheat, the gut of Nile tilapia were capable of fermenting all tested carbohydrate sources (highest for glucose and lowest wheat starch) and it was subsequently claimed that this would not only increase energy production but facilitate a less environmentally friendly intestine for pathogens. Amirkolaie *et al.* (2006) investigated the effects of high and low starch being either gelatinized or native, and it was demonstrated that gelatinized diets provided superior growth rates and faecal removal efficiency to Nile tilapia, and importantly, was also linked with the lower VFA production within the intestine and hence microbial fermentation.

Non-starch polysaccharides are categorized as cellulose (insoluble), pectic polysaccharides and non-cellulose polymers and fish lack the enzymes responsible for their digestion (e.g. β -glucanases and β -xylanases) as well as soluble non-starch polysaccharides potentially having antinutritional properties by increasing the viscosity of digesta thus reducing mineral and water absorption in tilapia (Francis *et al.* 2001; Hossain *et al.* 2003; Leenhouwers *et al.* 2007).

Research on non-starch polysaccharides is receiving increasing attention since NSP are present in higher amounts in tilapia diets with increasing use of plant-based ingredients. For example, for soybean meal and cottonseed meal, the reported soluble NSP content is 63 and 61 g kg⁻¹, respectively, with a total NSP of 217 and 283 g kg⁻¹, respectively (Sinha *et al.* 2011). In the case of soluble NSP, such as guar gum, this has been shown to increase the digesta viscosity of digesta thus reducing the emptying of the gut, and decreased mineral absorption (Leenhouwers *et al.* 2007). Similarly, Hossain *et al.* (2003) implicated non-starch polysaccharides within endosperms of the Sesbania seed, when included beyond 9.7%, to reducing the apparent digestibility of protein, lipid and energy to tilapia as well as their growth.

Due to the relative indigestible nature of some carbohydrate sources within plant-based ingredients, research has been conducted to determine whether pre-treating these ingredients would improve their utilization in tilapia (Ng *et al.* 2002; Belal 2008) or enzymatic activity (Li *et al.* 2009; Yigit & Olmez 2011). Palm kernel meal (PKM) was pre-treated with a commercially available enzyme or fermented with fungi (*Trichoderma koningii*) and it was found that enzyme treated palm kernel meal improved the growth and nutrient utilization in tilapia compared with raw palm kernel meal, while fermented palm kernel meal led to the worst performance (Ng *et al.* 2002). Ng *et al.* (2002) suggested that fungi metabolites may have acted as antinutrients in tilapia thus reducing their performance. In contrast, Belal (2008) demonstrated that using *T. koningii* to pre-treat date pits improved the nutrient digestibility of corn (as a carbohydrate source) in the diets of tilapia. Evidence was also presented to confirm that the fungi had a degradation effect based on the pit ultrastructure as well as detecting elevated sugar levels, with the potential release of other beneficial nutrients that otherwise would have been trapped within the plant cell walls.

Li *et al.* (2009) found that the exogenous supplementation of non-starch polysaccharides enzymes and phytase increased amylase activities in the hepatopancreas, but particularly in the intestine of tilapia. However, it is unclear whether this would translate into improved productivity or mitigate the potential negative effects of dietary non-starch polysaccharides, since growth was not measured. In another study, dietary cellulase had no significant effect on mitigating the high fibre content of dietary canola meal since there was no improvement in growth, nutrient digestibility or the body composition of tilapia (Yigit & Olmez 2011) and may be related to the relatively short transit time and the absence of specialized fermenting chambers in the gut for cellulase to act sufficiently (Saha *et al.* 2006). These findings are of particular note considering the exogenous supplementation of dietary enzymes, improved nutrient

utilization and growth in boiler chickens when fed higher carbohydrate diets (Kocher *et al.* 2002).

The mechanisms and reasons for better carbohydrate utilization in some fish over others is still not completely understood and may vary between different fish species (Leenhouwers *et al.* 2007), underscoring the importance of species-specific studies. What is known is that dietary NSP will continue to be an issue in tilapia nutrition and therefore research should continue into evaluating pre-treatments, including degradation methods (e.g. enzymes, heat, acidity) or the supplementation of beneficial bacteria or enzymes.

Vitamin and mineral requirements of tilapia

Vitamins and minerals are essential micronutrients that are involved in a wide variety of physiological processes necessary for the overall health of fish. Commercially prepared vitamin and mineral premixes are often added to the diets of tilapia, however, the composition of these premixes may be specific to one fish species and may not fulfill the optimal requirements of others. Moreover, the mineral and vitamin requirements may change depending on the culture conditions or dietary composition. In most cases, feed formulators are known to over supplement vitamins in aquafeeds to account for losses due to processing, storage and leaching. The known mineral requirements and signs of deficiencies are presented in Tables 3 and 4, respectively, while the known vitamin requirements and signs of deficiencies are presented in Tables 5 and 6, respectively.

Minerals are inorganic elements that include Mg, Na, K, Fe, Ca, P, Zn, Cr and Cu and play vital roles in enzyme function, including many of those responsible for digestion (e.g. lipase, alkaline phosphatase), gas exchange, acid/base balance (e.g. carbonic anhydrase) and osmoregulation (e.g. Na⁺/K⁺-ATPase). While many of these minerals can be obtained from brackish/marine environments, tilapia are mostly farmed in freshwater/low salinity waters and therefore dietary supplementation is essential for optimal health and productivity. The optimal mineral requirements of tilapia are influenced by the ionic composition of the culture water, the dietary mineral form and interactions with dietary protein or carbohydrates and may explain some of the discrepancies in research results among different studies. Quantitative mineral requirement studies are further complicated by the fact that some minerals are easily absorbed by the fish from the culture water and by contamination from containers and utensils used in laboratory analysis.

Mineral deficiencies or excesses can result in reduced performance and therefore careful considerations are important. This ultimately will help in designing diets containing the most appropriate mineral level and type for optimal growth or the selection of the most appropriate

Table 3 The recommended mineral requirements of tilapia

Mineral	Mineral form	Species and size (g)	Culture condition	Recommended level	Reference
Ca	Ca-sulphate	<i>O. aureus</i> (0.8)	FW (0 mg L ⁻¹ Ca)	7 mg kg ⁻¹	Robinson <i>et al.</i> (1987)
Ca	Ca-lactate	<i>O. niloticus</i> × <i>O. aureus</i> (0.5)	FW (27.1–33.3 mg L ⁻¹ Ca)	3.5–4.3 mg kg ⁻¹	Shiau and Tseng (2007)
Cr	Cr ₂ O ₃	<i>O. niloticus</i> × <i>O. aureus</i> (0.5)	FW	204 mg kg ⁻¹	Shiau and Shy (1998)
Cr	Cr-piccolinate	<i>O. niloticus</i> × <i>O. aureus</i> (1.3)	FW	Not required†	Pan <i>et al.</i> (2003)
K	KCl	<i>O. niloticus</i> × <i>O. aureus</i> (0.8)	FW (2.4 mg L ⁻¹ K)	2–3 g kg ⁻¹	Shiau and Hsieh (2001)
Mg	Mg-acetate (4H ₂ O)	<i>O. niloticus</i> (10.4)	FW (9.12 mg L ⁻¹ Mg)	0.59–0.77 mg kg ⁻¹ ‡	Dabrowska <i>et al.</i> (1989)
Mg	MgSO ₄	<i>O. aureus</i> (0.5)		5 g kg ⁻¹	Reigh <i>et al.</i> (1991)
Fe	Ferric citrate	<i>O. niloticus</i> × <i>O. aureus</i> (0.6)	FW (59.75 µg L ⁻¹)	150–160 mg kg ⁻¹	Shiau and Su (2003)
	Ferrous sulphate			85 mg kg ⁻¹	
P	Monobasic sodium phosphate	<i>O. aureus</i> (0.8)	FW (0.52 mg L ⁻¹ P)	5 g kg ⁻¹	Robinson <i>et al.</i> (1987)
Na	NaCl	<i>O. niloticus</i> × <i>O. aureus</i> (0.7)	FW	1.5 g kg ⁻¹	Shiau and Lu (2004)
			SW (32–34‰)	Not required	
NaCl	NaCl	<i>O. niloticus</i> × <i>O. aureus</i> (656)	FW	30–3.5 g kg ⁻¹	Cnaani <i>et al.</i> (2010)
Zn	Zinc sulphate (7H ₂ O)	<i>O. niloticus</i> (8.4)	FW	30 mg kg ⁻¹	Eid and Ghonim (1994)
Zn	Zinc sulphate monohydrate	<i>O. niloticus</i> (13.3)	FW (7.95 µg L ⁻¹ Zn)	79.5 mg kg ⁻¹	do Carmo e Sá <i>et al.</i> (2004)

FW, freshwater; SW, seawater.

†Suggested to be the result of a flow-through system preventing Cr accumulation.

‡Depends on the dietary protein level (lower protein requires less Mg).

Mineral	Role	Signs of deficiencies
Ca	Essential component in bone and scales for mineralization	Lower growth
Cr	Improves carbohydrate utilization	Lower growth
K	Essential for osmoregulation	Lower growth Depressed Na ⁺ /K ⁺ -ATPase activity
Mg	Aids in protein utilization	None reported
Fe	Essential component in blood	Lower growth Lower haematological values
P	Essential component in phospholipids, ATP and bone	Lower growth Lower P and Ca deposition in bones
Na and Cl	Major ion species in blood Essential for osmoregulation and acid/base balance	Lower growth Depressed Na ⁺ /K ⁺ -ATPase activity
Zn	Essential component in the enzymes carbonic anhydrase, alkaline phosphatase and alcohol dehydrogenase Essential component in cellular membranes Aids in protein and carbohydrate utilization	Lower growth and protein efficiency ratios Increased liver glycogen Increased moisture in muscle

†Refer to Table 3 for references.

Table 4 Role of minerals and signs of deficiencies in tilapia†

commercially made diets since these can contain varied mineral contents (Tacon & De 1983). It must also be noted that, generally, nutritional studies determining the optimal dietary mineral supplementation utilize purified diets, and therefore in practical situations, when fishmeal or other protein sources are used, these can inherently contain minerals or, in some cases when antinutritional factors are present, reduce their availability.

Major minerals present in water include Na, Cl, K, Mg and Ca and interpreting these dietary requirements to tilapia will depend greatly on the ionic composition of the culture water (Shiau & Lu 2004; Shiau & Tseng 2007; Cnaani *et al.* 2010). Generally, those cultured in brackish/seawater may not require the above mentioned minerals, while conversely, those under hypo-osmotic conditions will require dietary minerals to maintain ionic

Table 5 Dietary vitamin requirement of tilapia. Units are mg kg⁻¹ of diet unless otherwise stated

Vitamin	Vitamin form	Tested inclusion levels	Species and size (g)	Recommended level	Reference
Water soluble					
Choline	Choline chloride	0–4000	<i>O. niloticus</i> (3.1–3.4)	3000†	Kasper <i>et al.</i> (2000)
Choline	Choline chloride	0–2000	<i>O. niloticus</i> × <i>O. aureus</i> (0.6)	1000‡	Shiau and Lo (2000)
Thiamin (B ₁)	Thiamin hydrochloride	0–16	<i>O. niloticus</i> (3.6–5.3)	3.5	Lim <i>et al.</i> (2011b)
Riboflavin (B ₂)	Riboflavin	0–48	<i>O. aureus</i> (0.7)	6	Soliman and Wilson (1992a)
Riboflavin (B ₂)	Riboflavin	0–40	<i>O. mossambicus</i> × <i>O. niloticus</i>	5 (in seawater)	Lim <i>et al.</i> (1993)
Niacin (B ₃)	NP§	0–200	<i>O. niloticus</i> × <i>O. aureus</i> (2.2)	26¶ 121**	Shiau and Suen (1992)
Pantothenic acid (B ₅)	Calcium d-pantothenate	0–40	<i>O. aureus</i> (0.7)	10	Soliman and Wilson (1992b)
Pantothenic acid (B ₅)		0–20	<i>O. niloticus</i> (13.7)	10	Ayyat <i>et al.</i> (2011)
Pyridoxine (B ₆)	Pyridoxine hydrochloride	0–100	<i>O. niloticus</i> × <i>O. aureus</i> (0.7)	1.7–9.5†† 15.0–16.5‡‡	Shiau and Hsieh (1997)
Biotin (B ₇)	Biotin	0.05–5.0	<i>O. niloticus</i> × <i>O. aureus</i> (1)	0.06	Shiau and Chin (1999)
Folic acid (B ₉)	Folic acid	0–20	<i>O. niloticus</i> × <i>O. aureus</i> (0.4)	0.82	Shiau and Huang (2001)
Folic acid (B ₉)	Folic acid concentrate	0–6	<i>O. niloticus</i> (14.5)	0.5–1.0	Barros <i>et al.</i> (2009)
Vitamin B ₁₂	Vitamin B ₁₂	0–200 µg kg ⁻¹	<i>O. niloticus</i> × <i>O. aureus</i> (1)	Not required	Shiau and Lung (1993)
Vitamin C	Ascorbic acid	0–400 mg 100 kg ⁻¹	<i>O. niloticus</i> × <i>O. aureus</i> (1)	125 mg 100 kg ⁻¹	Soliman <i>et al.</i> (1994)
Vitamin C	C2S	0–120	<i>O. niloticus</i> × <i>O. aureus</i> (1.5)	41–48	Shiau and Hsu (1995)
	C2MP-Mg	0–120		37–42	
	Ascorbic acid	0–120		80	
Vitamin C	C2MP-Na	0–120	<i>O. niloticus</i> × <i>O. aureus</i> (1.2)	63.4	Shiau and Hsu (1999)
	C2MP-Mg	0–120		40.5	
Inositol	myo-inositol	0–800 mg kg ⁻¹	<i>O. niloticus</i> (5.8)	100–400 mg kg ⁻¹	Peres <i>et al.</i> (2004)
Inositol	myo-inositol	0–1253 mg kg ⁻¹	<i>O. niloticus</i> × <i>O. aureus</i> (0.5)	400 mg kg ⁻¹	Shiau and Su (2005)
Lipid soluble					
Vitamin A	NP	0–40000 IU kg ⁻¹	<i>O. niloticus</i> (11.4)	5000 IU kg ⁻¹	Saleh <i>et al.</i> (1995)
Vitamin A	Retinyl acetate	0–50000 IU kg ⁻¹	<i>O. niloticus</i> × <i>O. aureus</i> (1.6)	5850–6970 IU kg ⁻¹	Hu <i>et al.</i> (2006)
Vitamin A	Vitamin A acetate	500–8000 IU kg ⁻¹	<i>O. niloticus</i> (7.7)	Not required§§	Guo <i>et al.</i> (2010)
β-carotene	β-carotene	0–200	<i>O. niloticus</i> × <i>O. aureus</i> (0.5)	28.6–44¶¶	Hu <i>et al.</i> (2006)
Vitamin D	Cholecalciferol D ₃	0–2100 IU kg ⁻¹	<i>O. niloticus</i> × <i>O. aureus</i> (0.8)	374.8 IU kg ⁻¹	Shiau and Hwang (1993)
Vitamin E	Tocopherol-acetate	0–100	<i>O. niloticus</i> (7.5)	20–100	Eleraky <i>et al.</i> (1995)
Vitamin E	α-tocopheryl acetate	0–700	<i>O. niloticus</i> × <i>O. aureus</i> (0.7)	63–206	Huang <i>et al.</i> (2003)

†With 0.5 g total sulphur amino acids 100 g⁻¹ protein.‡With 0.53 g total sulphur amino acids 100 g⁻¹ protein.

§Not provided.

¶When glucose was used as a carbohydrate source.

**When dextrin was used as a carbohydrate source.

††At a protein level of 28%.

‡‡At a protein level of 36%.

§§Speculated to be the result of high vitamin A levels in the cod liver oil used as the dietary lipid source.

¶¶When dietary vitamin A was present at 84 IU kg⁻¹.

homeostasis. For example, when freshwater reared tilapia were fed with diets containing different NaCl levels, higher growth rates corresponded with lower gill Na⁺/

K⁺-ATPase activity (Shiau & Lu 2004), likely linked with less energy being channeled for hyper-osmoregulation. However, with regard to Mg, either deficiencies in water

Vitamin	Role	Signs of deficiencies
Choline	Cellular repair	Lower growth
Thiamin	Coenzyme to thiamin pyrophosphate	Anorexia
	Essential for various metabolic pathways	Instability/lethargy
		Lower haematocrit values
		Lower feeding efficiency and growth
Riboflavin	Essential for erythrocyte glutathione reductase	Anorexia and reduced feed intake
	Essential for liver D-amino acid oxidase	Nervous symptoms
		Fin erosion
		Cataracts
		Changes in normal body color
		Lower liver D-amino acid oxidase
		No histopathological changes
Niacin	Precursor for coenzymes for carbohydrate metabolism	Skin/fin lesions
		Haemorrhages
		Snout deformations
Pantothenic acid	Precursor for coenzyme A for carbohydrate metabolism	Gill lesions
		Reduced food intake
Pyridoxine	Essential for protein/AA metabolism	Anaemia
		Lower growth
		Lower hepatic alanine aminotransferase
Folic acid		Lower growth
		No outward signs
		No change to haematological values
Vitamin C	Important for immunity	Lower growth
	Important for collagen synthesis	
	Antioxidant properties	
Inositol	Structural component in phospholipids of cell membranes	Lower growth
	Involved in transduction pathways	Higher muscle/liver lipid
Vitamin A	Vision	Haemorrhages
	Cell growth	Exophthalmia
		Oedema
		Reduced mucous secretion
		Lower growth and survival
Vitamin D	Possible role in Ca and P bone mineralization	Lower growth
Vitamin E	Antioxidant properties	Lower growth
	Important roles in reproduction	

†Refer to Table 5 for references.

Table 6 Role of vitamins and signs of deficiencies in tilapia†

borne Mg ($<5 \mu\text{M}$) or dietary Mg (1.5 mmol kg^{-1}) had no significant effect on the growth or gill Na^+/K^+ -AT-Pase activity of Mozambique tilapia (Bijvelds *et al.* 1997). Similarly, dietary Mg deficiencies had no significant growth effect on Nile tilapia, although inclusion rates between 0.59 and 0.77 g kg^{-1} have been reported to provide optimal growth and protein utilization (Dabrowska *et al.* 1989). A combination of a low dietary protein content (24%) and high Mg of 3.10 g kg^{-1} was excessive since this significantly reduced the growth of Nile tilapia, and was suggested to be the result of under-utilized Mg during protein metabolism leading to toxic

effects (Dabrowska *et al.* 1989). Therefore caution should be used when performing mineral supplementation when culturing tilapia in freshwater and utilizing low protein diets. While phosphorus is a major mineral in seawater, no studies have indicated a change in P requirements at different salinities. Nevertheless, dicalcium phosphate (DCP) is often added to tilapia diets since this has been shown to be essential for optimal growth and bone mineralization (Robinson *et al.* 1987) and, as previously mentioned, is likely to become a limiting nutrient when dietary plant-based ingredients are used (Viola *et al.* 1988; Furuya *et al.* 2004; Goda *et al.* 2007).

Trace minerals include Cr, Zn, Cu and Fe, and the majority of studies have all shown that these supplementations in the diet improves the growth of tilapia (Shiau & Chen 1993; Shiau & Lin 1993; Eid & Ghonim 1994; Shiau & Liang 1995; Shiau & Shy 1998; Shiau & Su 2003; Zhao *et al.* 2011) although, with the exception of Fe, the exact role of these in tilapia metabolism still requires more investigation. Aside from being an essential component to blood cells, dietary Fe supplementation may explain the improved growth, feeding efficiency and haematological values of tilapia, although it was shown that ferric citrate was 49% less available for tilapia than ferrous sulphate (Shiau & Su 2003). Li *et al.* (2007) reported that dietary supplementations of Cu, Fe and Zn to tilapia hybrids significantly increased amylase activities in the hepatopancreas and intestine while lipase activity significantly increased in the intestine, indicating that starch and lipid digestion would be improved, respectively. With regard to Zn, in the form of ZnSO₄, this has been shown to significantly decrease liver glycogen levels and to increase liver glucose-6-phosphate dehydrogenase (G6P-DH) activity, which was suggested to improve carbohydrate digestion, thus explaining the higher growth in tilapia (Zhao *et al.* 2011). Interestingly, in the same study, despite ZnMet being substantially less effective in reducing liver glycogen levels or enzymatic activity in tilapia, the growth performance and feeding efficiency of tilapia fed this zinc form were similar to those fed ZnSO₄ supplementations, which may indicate other mechanisms at work in maintaining high growth rates. Reports on dietary chromium improving carbohydrate utilization and growth in tilapia have been contradictory. It has been shown that inorganic sources of chromium improved the growth performance of tilapia, suggested to be the result of enhanced carbohydrate utilization possibly due to chromium acting as a cofactor in the activity of insulin (Shiau & Chen 1993; Shiau & Lin 1993; Shiau & Liang 1995; Shiau & Shy 1998). However, organically bound chromium picolinate (Cr-Pic) dietary supplementation caused no improvement to the growth, feeding efficiency, carbohydrate utilization (either from glucose or dextrin sources) or enzymatic activities associated with lipogenesis and carbohydrate digestion (Pan *et al.* 2003). Pan *et al.* (2003) suggested that these discrepancies may be the result of rearing conditions where an accumulation of chromium may have occurred in the recirculating system used in previous studies, whereas in their experiment, a flow-through system was used thus preventing any such accumulation in the culture water. Clearly more research is required regarding the role of chromium in nutrient utilization, and since it is known that tilapia may experience hyperglycaemia when consuming high carbohydrate diets, studies on examining blood glucose levels as well as

fecal composition at different dietary chromium supplementation may be worthwhile.

Vitamins are organic nutrients that are essential for a wide variety of processes, and include aiding nutrient utilization, improving immunological responses, providing antioxidant protection and maintenance of other basic physiological processes. Vitamins are broadly categorized as either water soluble or lipid soluble, which will have implications to longer term storage in both the diets of tilapia and in tilapia tissues.

Water soluble vitamins include inositol, choline, vitamin B groups, vitamin C and biotin. Both inositol and choline have been described as 'vitamin-like' which are essential components to phospholipids. Peres *et al.* (2004) showed that tilapia had no specific requirement for inositol but since inositol plays an important role in lipid metabolism, it was suggested that 100–400 mg kg⁻¹ diet should be added to reduce lipid accumulation in the muscle and liver. Similarly, Shiau and Su (2005) also recommended a supplementation of 400 mg kg⁻¹ diet, based on improved growth of hybrid tilapia and noted that intestinal bacteria were likely insufficient to meet the requirement. Choline is important in S-adenosylmethione synthesis (essential for the growth and repair of cells) and the dietary requirement of choline can reportedly be satisfied by synthesis from the total sulphur containing amino acids (TSAA) when provided in abundance (Kasper *et al.* 2000; Shiau & Lo 2000). Interestingly, however, even though dietary TSAA were similar at approximately 0.5 g 100 g⁻¹ TSAA, there was a three-fold higher recommended choline requirement level in the study by Kasper *et al.* (2000) compared with those reported by Shiau and Lo (2000). Moreover, the provided dietary TSAA was lower than optimal levels for tilapia (Santiago & Lovell 1988), thus ensuring minimized *in vivo* choline synthesis. It is unlikely that this discrepancy is the result of fish weight differences since older fish were used in the study by Kasper *et al.* (2000), and the requirements for choline reportedly decrease with increasing age (Woodward 1994). Nevertheless, Kasper *et al.* (2002) found that for tilapia, dietary betaine can completely replace choline without significantly reducing their weight, with a choline: betaine ratio of 10: 90 providing the best growth. Moreover, since betaine may provide additional nutritional benefits, such as improved diet palatability and flavour, it was therefore suggested that betaine be used rather than choline, particularly since total sulphur containing amino acids often become limiting when using plant-protein sources (Kasper *et al.* 2000; Shiau & Lo 2000). Recently, in choline-free diets, a dietary betaine level of 12.5 g kg⁻¹ diet was reported to be optimal for growth and improved enzymatic activity in the liver for tilapia (Luo *et al.* 2011). However, caution should be used with these nutrients

since a level beyond 12.5 g kg^{-1} betaine tended to reduce growth performance (Luo *et al.* 2011) and similarly Kasper *et al.* (2000) noted that if choline is provided in excess of 4 g kg^{-1} , growth can be depressed.

Vitamin B is a group of vitamins that play essential roles in fish metabolism and erythrocyte production and while these can be found naturally in plants or animal sources, supplementation is common (Shiau & Suen 1992; Soliman & Wilson 1992a,b; Ayyat *et al.* 2011; Lim *et al.* 2011a). However, it has been shown that both B_9 (folic acid) and B_{12} are not required for Nile tilapia (Lovell & Limsuwan 1982; Shiau & Lung 1993), likely due to the capacity of gut bacteria (e.g. bacteroides type A for B_{12} synthesis) to synthesize these nutrients (Lovell & Limsuwan 1982; Sugita *et al.* 1991; Shiau & Lung 1993; Shiau & Huang 2001; Barros *et al.* 2009). In contrast, Shiau and Huang (2001) reported vitamin B_9 to be a growth promoter in a dose-dependent manner at the tested levels of 0.3 and 20 mg kg^{-1} diet, with 0.82 mg kg^{-1} diet being recommended.

Currently, B_1 (thiamin), B_2 (riboflavin), B_3 (niacin), B_5 (pantothenic acid), B_6 (pyridoxine) and B_7 (biotin) have all been found to be essential for optimal growth and nutrient utilization in tilapia and deficiencies include poor growth, neurological symptoms, lesions and haemorrhages (Shiau & Suen 1992; Soliman & Wilson 1992a,b; Lim *et al.* 1993, 2011a; Shiau & Hsieh 1997; Shiau & Chin 1999; Ayyat *et al.* 2011). Interestingly, Lim *et al.* (2011a) noted that vitamin B_1 deficiencies were brought on faster when Nile tilapia were fed diets with higher dietary carbohydrate levels as this vitamin is likely utilized at accelerated rates when higher dietary carbohydrates were present. A similar mechanism may also explain a higher vitamin B_3 requirement when different dietary carbohydrate sources were used (Shiau & Suen 1992). Finally for vitamin B_6 , higher levels were needed at higher dietary protein levels, suggested as a result of being involved with the hepatic alanine aminotransferase activity and likely other protein/amino acid enzymatic pathways (Shiau & Hsieh 1997). Investigations on whether similar trends apply to vitamin B_2 are warranted.

Vitamin C has been shown to have a sparing effect on vitamin E, is essential to collagen synthesis and improves growth and immunological resistance of tilapia to pathogenic bacteria (Shiau & Hsu 1995; Ibrahim *et al.* 2010). Similarly to vitamin E, vitamin C has antioxidant properties, however, since it is lipid insoluble it is less persistent, particularly L-ascorbic acid (Shiau & Hsu 1995). It has been suggested that both L-ascorbyl-2-monophosphate (C2MP) and L-ascorbyl-2-sulphate (C2S) are more stable and both have been demonstrated to exert the same antiscorbutic activity, although C2MP was more effectively stored in tilapia (Shiau & Hsu 1995). Nevertheless, considering these are lipid insoluble, investigations on their protective effects

against lipid peroxidation in comparison with vitamin E need to be elucidated. With regard to C2MP, it was found that C2MP-Na was only 85% as effective as C2MP-Mg, in terms of growth and feeding efficiency, as well as causing higher ascorbate levels in the tilapia liver (Shiau & Hsu 1999). Further, Ibrahim *et al.* (2010) demonstrated that when supplied as L-ascorbyl-2-polyphosphate at 500 mg ascorbic acid equivalent kg^{-1} diet, this was shown to significantly improve the resistance of tilapia to the bacterial pathogen, *Aeromonas hydrophila*, as well as act as a growth promoter to tilapia. However, it was recommended that at this level, the duration of dose should not exceed 1 month since the relative level of protection declines with time (Ibrahim *et al.* 2010).

The lipid soluble vitamins include vitamin A, D, E and K. Vitamin A is well known to be important for vision, metabolic processes and growth in fish, and therefore dietary supplementation in the diets of tilapia is essential (Saleh *et al.* 1995). As noted by Saleh *et al.* (1995), deficiencies or excessive amounts of vitamin A can lead to reduced survival, growth and condition factor. Two dietary factors that may influence the required supplemented vitamin A levels include the lipid source and β -carotene levels (Hu *et al.* 2006; Guo *et al.* 2010). Guo *et al.* (2010) reported that no dietary supplementation of vitamin A was required for Nile tilapia and suggested that this was the result of utilizing cod liver oil as the lipid source which inherently contains high vitamin A. It has been demonstrated that tilapia hybrids can synthesize vitamin A from β -carotene, consequently leading to substantially lower required vitamin A supplementation (Hu *et al.* 2006). Therefore, considering certain plant-based oils, particularly crude palm oil, are naturally rich in β -carotene (Ng & Gibon 2011), future directions may include researching the possibility of partial or complete replacements for added synthetic vitamin A in tilapia diets as well as elucidating the conversion efficiency of these endogenous carotenoids present in plant oils.

Vitamin D_3 has been shown to be essential for optimal growth in tilapia hybrids (Shiau & Hwang 1993) and blue tilapia (O'Connell & Gatlin 1994). The primary role of vitamin D_3 is to improve calcium intestinal absorption in terrestrial animals, however, its effects on Ca and P bone mineralization in tilapia have been contradictory (Shiau & Hwang 1993; O'Connell & Gatlin 1994; Rao & Raghuramulu 1999). In long term studies, dietary vitamin D_3 significantly increased calcium bone levels in a dose-dependent manner for tilapia hybrids (Shiau & Hwang 1993), however, this was not demonstrated with blue tilapia (O'Connell & Gatlin 1994). Moreover, when Rao and Raghuramulu (1999) injected Mozambique tilapia with vitamin D_3 and metabolites [i.e. 25-OH-D_3 and $1,25\text{-(OH)}_2\text{D}_3$], this had no significant effect on blood calcium

levels, gill calcium binding protein (CaBP) activity or Ca/P uptake within 1–4 days. Further research is required fully to explain these discrepancies.

Among the vitamins, a great deal of attention has been focused on dietary vitamin E supplementation in tilapia diets due to the powerful antioxidant properties, to both pre- and post-harvest fish, the potential role in modulating fatty acid metabolism and is essential for optimal growth (Eleraky *et al.* 1995; Huang *et al.* 2003; Huang & Huang 2004; Wang *et al.* 2006; Lim *et al.* 2009). Vitamin E is lipid soluble and has four naturally occurring isomers (α , β , γ , δ) of tocopherol (T) and tocotrienol (T3), although synthetic α -tocopherol acetate is commonly used in fish diets (reviewed by Hamre 2011). Since the primary role of vitamin E is as an antioxidant, requirements in tilapia will depend on the fatty acid composition of the diets, the levels and oxidative state of the lipids, oxidative stress of the fish and vitamin E isomer type. Therefore quantifying the requirements is somewhat difficult. For example, since LC-PUFA are more susceptible to lipid peroxidation, higher vitamin E supplementation may be useful in providing greater oxidative stability to the diets while protecting the fish from associated free radicals (Huang & Huang 2004). It has also been shown that vitamin E deposition extends to all major tilapia organs, albeit there was organ isomer specificity, thus providing broad oxidative protection (Wang *et al.* 2006). T3 is increasingly being investigated since there are indications

of a higher oxidative activity than T, potentially having synergistic activities with other isomers as well as being naturally present at higher concentrations in certain plant oils, particularly palm oil (Kamat *et al.* 1997; Wang *et al.* 2006). In contrast to vitamin C, an excess supply of dietary vitamin E supplementation (of 240 mg kg⁻¹) and in their reported optimal range (of 50–200 mg kg⁻¹) had no effect on the resistance of tilapia to *Edwardsiella tarda* (Kim *et al.* 2003) or *Streptococcus iniae* challenge (Lim *et al.* 2009).

In a recent review, Krossøy *et al.* (2011) discussed the role and mechanisms of vitamin K in fish nutrition and stated that, despite being essential for optimal growth and physiological processes, there is a general lack of requirement data for fish, including tilapia. There are two types of naturally occurring vitamin K, phyloquinone (vitamin K₁) and menaquinone (vitamin K₂), which are synthesized by plants and microbes, respectively, although the most commonly used in fish feeds is the synthetic menadione (vitamin K₃) (Krossøy *et al.* 2011). However, there are indications vitamin K₃ is much more unstable than vitamin K₁ and K₂, thus making quantitative recommended levels in fish nutrition more difficult, while in practical feeds, a substantial amount can be lost during prolonged storage (Marchetti *et al.* 1999). Currently the recommended vitamin K levels range between 0.1 and 20 mg kg⁻¹ for Atlantic salmon and haddock, respectively. However, as noted by Krossøy *et al.* (2011) this may not be due to species-specific

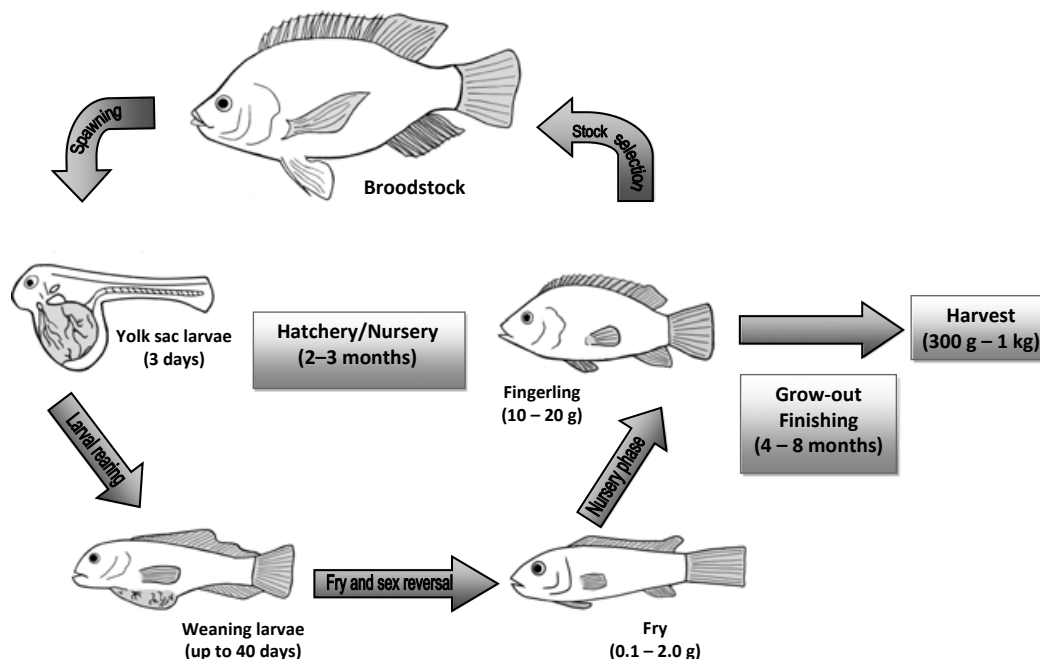


Figure 4 Culture cycle of tilapia from broodstock to market size (fish stages not drawn to scale). The dietary nutritional content and feeding management will greatly influence the duration of each stage.

differences but rather variation in the experimental conditions, type of food, age (e.g. transfer from broodstock to fry) and type of vitamin used.

Practical applications of feeding management and nutrition throughout the life cycle

It is known that the nutritional requirements can change depending on the life stage of tilapia (Fig. 4), while different feeding regimes and feed formulations can also have a significant effect (Fig. 5). Since the focus in the previous sections mostly referred to the nutritional requirements of tilapia fingerlings and at grow-out stages, this section will discuss the different nutritional requirements as well as the most appropriate feeding management strategies for tilapia broodstock and their fry to maximize productivity. This will be followed by a discussion on practical feed formulations, implications of different farmed tilapia strains to nutrient utilization and finally on finishing diets to improve fillet quality at harvest.

Nutritional requirements and feeding management for broodstock

Tilapia from the genus *Oreochromis* and *Sarotherodon* are mouthbrooders that have been described as having asynchronous spawning cycles as well as low fecundity that decreases over time, and these characteristics are viewed as major limitations to the expansion of commercial tilapia aquaculture (Bhujel 2000; Getinet 2008). It is well known that nutrition plays a vital role in fish fecundity (Izquierdo *et al.* 2001) and while it has been stated that commercial feeds designed specifically for tilapia broodstock are unavailable due to a perceived lack of demand (Bhujel

2000), this appears to be slowly changing over the past decade. Therefore to select the most appropriate diet or ingredients, it is important to understand the different nutritional requirements of broodfish, compared with other tilapia life stages, which will also have implications to offspring quality. Furthermore, with appropriate feeding management, the production of tilapia fry can be manipulated to coincide with seasonal market demands.

In contrast to the grow-out of tilapia, higher dietary protein levels in broodfish diets are more crucial for productivity (Santiago *et al.* 1985; Gunasekera *et al.* 1995, 1996, 1997; Bhujel *et al.* 2001; El-Sayed *et al.* 2003; El-Sayed & Kawanna 2008; Lupatsch *et al.* 2010). For example, feeding tilapia fry (0.3 g) a dietary protein level between 10% and 17% resulted in the fish never reaching sexual maturity (Gunasekera *et al.* 1995). Moreover, fry receiving 25% protein had slow oocyte maturation (Gunasekera *et al.* 1995) and for sexually mature broodfish, diets of 10% protein caused none of their eggs to be fertilized (Gunasekera *et al.* 1996). Based on various studies in different culture systems, dietary protein levels of 30–40% in the feeds of tilapia broodfish are reported to optimize fecundity by decreasing the age of sexual maturity and/or increasing seed production (El-Sayed *et al.* 2003; El-Sayed & Kawanna 2008; Lupatsch *et al.* 2010). It should also be noted that providing higher dietary protein levels was shown to be more important at higher salinities since a combination of low protein levels of 25% at salinities of 7‰ and 14‰, resulted in longer spawning intervals, egg hatch duration and yolk sac absorption as well as reducing overall fecundity compared with the same protein level in freshwater (El-Sayed *et al.* 2003). Feeding costs can be reduced since alternating high (40%) and low (25%) protein diets each day or at three and two consecutive days, respectively, to female tilapia

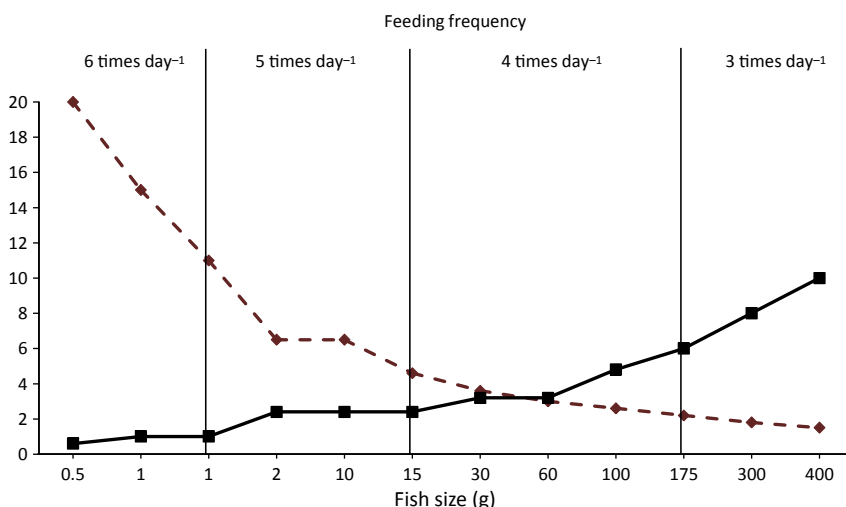


Figure 5 The recommended feeding rate (% body weight), pellet size (mm) and feeding frequency (times/day) as a function of fish size (g). The minimum dietary protein level for tilapia is 40, 30, 25 and 20% at fish weights of <0.5–2 g, 10–30 g, 60–175 g and 300–400 g, respectively. Data modified from Creswell (2005). —◆—, Feeding rate; —■—, Pellet size (mm).

broodstock led to no significant difference to fry production (Santiago & Laron 2002).

It has been shown that commercially available catfish pellets (of 3 and 5 mm with a crude protein content of 30% and 25%, respectively), often used in tilapia farming in Thailand, increased the seed production from Nile tilapia broodfish within a hapa-in-pond system over feeds with crude protein of 15.5% (Bhujel *et al.* 2001). Although some studies on broodstock fecundity utilized both soybean meal and fishmeal as a protein source (e.g. Santiago *et al.* 1985; Lupatsch *et al.* 2010), direct comparisons between fishmeal and alternative protein sources, as either partial or complete replacements, have not been investigated to date. Nevertheless, to maximize the use of protein for reproductive investment, rather than being catabolized for energy, the optimal protein to energy (P/E) ratios has been studied. Both El-Sayed and Kawanna (2008) and Lupatsch *et al.* (2010) recommended a similar protein to energy (P/E) ratio of 21.6–23.6 mg CP KJ g⁻¹ and 20.5 mg CP KJ g⁻¹, respectively, when high dietary protein levels were used.

Feed additives have also been investigated on tilapia fecundity where Therigon[®] (Adwia Co, El Oubor, Egypt), consisting of α -amino-p-hydroxyhydrocinnamic acid and Nuvisol Hatch P[®] (Newtrix Co, Lille, Belgium), consisting of various B vitamins, L-carnitine and progesterone, at 1 g kg⁻¹ of diet fed to female tilapia significantly increased the number of eggs/fish (Abdelhamid *et al.* 2010). However, due to the high cost of these additives, Abdelhamid *et al.* (2010) suggested that the most economical practice was the use of 0.5 g Therigon[®] kg⁻¹ of diet. Plants oils, which are relatively cheaper and readily available, have been reported to produce similar or even better results to the reproductive performance of tilapia than the sole use of fish oil in broodfish diets (Santiago & Reyes 1993; El-Sayed *et al.* 2005; Hajizadeh *et al.* 2008; Ng & Wang 2011).

Santiago and Reyes (1993) tested cod liver oil (CLO), soybean oil, corn oil, coconut oil or a CLO + CO blend (1:1) to fishmeal-based diets on the reproductive performance of Nile tilapia. It was shown that cod liver oil was inferior to all tested lipid sources and the addition of soybean oil in the broodfish diets led to the highest number of spawned fish (100% spawning), spawning cycle as well as good fry production per fish (Santiago & Reyes 1993). In another study comparing dietary fish oil, soybean oil or fish oil + soybean oil blends, all diets (1:4 fishmeal to soybean meal ratio) produced similar broodstock fecundity as well as spawning interval and egg diameter, although at higher salinities of 7‰ and 14‰, the sole use of dietary soybean oil was less effective (El-Sayed *et al.* 2005). A more long term study evaluated the fecundity over three consecutive spawnings, and from the beginning of exogenous feeding of Nile tilapia with a commercial diet designed for trout as well as experimental diets containing cod liver oil, palm oil

(PO) or a combination of PO + CLO (9:1), with soybean meal being the sole protein source (Hajizadeh *et al.* 2008). It was demonstrated that tilapia fed the CLO-based diets experienced high mortalities and no spawned fish, while those fed the palm oil and PO + CLO had higher fecundity than those fed the commercial diet (Hajizadeh *et al.* 2008). This is a particularly important finding considering these diets were fed throughout the life of tilapia, indicating that feeding costs to tilapia broodstock can be substantially reduced with the use of palm oil. The beneficial effects of PO in tilapia broodstock diets was further confirmed when over at least five consecutive spawnings, Ng and Wang (2011) observed that fish fed diets (1:2.6 fishmeal to SBM) supplemented with CPO or CPO + fish oil (1:1) had a shorter first spawning and inter spawning intervals as well as a higher percentage of spawning females, hatching rates and total egg production compared with broodfish fed on the fish oil-based diets. While the use of palm oil products enhanced broodstock reproductive performance, there are also indications of additional nutritional benefits passed onto the newly hatched fry. Crude palm oil is naturally rich in vitamin E, consisting of both tocopherols and tocotrienols, and it was observed that broodstock fed diets containing crude palm oil passed on vitamin E to the fry thus enhancing their resistance to oxidative stress (unpublished results). Based on these findings, plant-based oils can totally replace fish oil and, to date, the use of crude palm oil or blends with fish oil appears to be the best lipid source for optimal broodstock fecundity and nutritive benefits to their offspring. Generally, studies on lipid sources used for tilapia broodstock reproduction are incorporated at between 5% and 10% of diet, however, to date, the optimal lipid levels for tilapia broodstock have yet to be determined.

For the most effective use of feeds, feeding management plays a crucial and influential role. Interestingly, it has been shown that generally lower feeding rates to tilapia broodfish are more effective, particularly from a cost perspective (Mironova 1977; Siddiqui *et al.* 1997; Bhujel *et al.* 2007; Tsadik & Bart 2007; Matsiko *et al.* 2010). Matsiko *et al.* (2010) noted that it was a common practice on some tilapia farms in Africa to feed broodstock with commercial feeds at rates as high as 5% body weight (BW) since this was presumed to increase fecundity. However, it was shown that feeding rates between 0.5% and 5% BW led to no significant difference in seed production and, when taking into consideration cost versus production, there was an over 12-fold decrease in the profit index when a 5% BW feeding regime was used compared with 0.5% BW (Matsiko *et al.* 2010). Siddiqui *et al.* (1997) also recommended a feeding rate of 0.5–1% BW for hybrid tilapia, since there was more production of eggs g⁻¹ of fish with more energy channeled to reproductive output rather than towards somatic

growth. A similar suggestion was made by Lupatsch *et al.* (2010) after quantifying the metabolic rates of both energy and protein investment for Nile tilapia, and moreover when fed to satiation, their feed intake naturally reduced to approximately 0.6% BW. It should be noted, however, that restricted feed for prolonged periods may reduce spawning intervals and total fecundity (Lupatsch *et al.* 2010) and therefore more research into cyclic feeding regimes should be investigated. Using a lower feeding rate for broodfish will likely provide a variety of benefits including lower feeding costs and minimizing nitrogenous waste production, which is particularly beneficial to reduce fouling in hapa systems used for tilapia breeding (Bhujel 2000). Moreover, if feed is provided in excess leading to higher growth rates of broodstock, this may become undesirable since fish over 250 g are sometimes discarded due to the belief that they are more susceptible to disease and sub-optimal environmental conditions in addition to having higher feed intake rates, lower relative fecundity and being more difficult to handle when removing eggs for subsequent incubation (Bhujel 2000; Bhujel *et al.* 2001).

Continual seed production is sometimes undesirable due to seasonal fry demands. In this situation, deliberately stunting the growth of fingerlings (serving as future broodstock) by reducing feeding rates to 1% BW at high stocking densities (e.g. >5000 fingerlings m^{-2} then 80 fish m^{-2}) is sometimes performed to save both space and feeding costs (Bhujel *et al.* 2007). While this can prolong sexual maturity, Bhujel *et al.* (2007) investigated the fecundity implications of stunted broodfish (age of 13 months) compared with unstunted broodfish (age of 4 months) at different feeding rates. It was observed that changing to a feeding rate of 3% BW led to a higher overall seed output for stunted broodstock than their unstunted counterparts, although feeding rates of 1% and 2% BW were lower and similar, respectively, for stunted broodstock (Bhujel *et al.* 2007). Bhujel *et al.* (2007) suggested that the rapid growth of stunted broodstock, exceeding that of normal fish when fed at 3% BW, likely caused a higher clutch size of eggs for the larger fish.

Nutritional requirements and feeding management for fry (0.1–2 g)

When tilapia larvae hatch, there is an initial reliance on their egg yolk sac for nutrition. However, this quickly becomes absorbed in approximately 3 days and therefore it becomes necessary to wean the larvae onto artificial diets (Fig. 4). Due to the relatively small size, and therefore mouths, starter diets in the form of powder or crumble (<0.06 mm) are often used. Moreover with their relatively fast growth rates, it has been recommended to provide high protein diets (40–50%), frequently (six times per day) at 30–45% BW or to

satiation (Fig. 5). It was demonstrated that, among three different protein levels of 25%, 35% or 45%, the highest dietary protein significantly enhanced the growth, increased feed intake and improved the feeding efficiency of Nile tilapia fry compared with those fed the lower dietary protein levels (Abdel-Tawwab *et al.* 2010). In terms of feeding frequency, MacIntosh and De Silva (1984) demonstrated that 12% and 24% BW provided optimal survival, growth and FCR to tilapia hybrid fry (when at an optimal stocking density of 2–12 fry L^{-1}), which was suggested to be linked with reduced cannibalism as well as supporting their fast growth. Further, using an optimal stocking density of 5 fry L^{-1} and dietary protein level of 40%, optimal feeding rates were evaluated at 10–35% BW and to satiation on the growth performance and feeding efficiency to Nile tilapia fry (El-Sayed 2002). It was found that feeding rates of 10% caused over a two-fold decrease in growth and a significantly lower survival rate (60%) of the fish compared with those fed 35% BW and to satiation with survival rates of 83% and 96%, respectively (El-Sayed 2002).

Consistently feeding high protein diets may not always be the best practice from a cost-effectiveness standpoint. De Silva and Perera (1984) noticed a rhythmicity to apparent protein and dry matter digestibility of red hybrid tilapia fry (1.0–1.3 g) and speculated that mixed feeding schedules of utilizing diets with different protein levels may reduce feeding costs without compromising productivity. This was subsequently tested in a later study using two size groups of tilapia (170–213 mg and 1.01–1.07 g), and it was indeed demonstrated that a feeding schedule of 18% dietary protein for two consecutive days followed by 30% dietary protein for three consecutive days led to similar growth rates of tilapia fry compared with those only fed a higher protein of 30% throughout (De Silva 1985). This led to a cost savings of 20% from fishmeal (De Silva 1985) and later studies utilizing diets containing soybean meal + fishmeal (near 1:1) or alternative protein mixes confirmed a similar feeding schedule to juvenile tilapia (Patel & Yakupitiyage 2003; El-Husseiny *et al.* 2008) indicating this cost saving practice applies to diets containing fishmeal or alternative protein sources. In addition, the concept of feeding economically optimal dietary concentrations of protein based on diminishing returns functions has also been advocated (De Silva *et al.* 1989). To maximize profits, it was calculated that a dietary protein level of approximately 28% was more cost-effective than at 36% (De Silva *et al.* 1989).

Since male tilapia grow faster than females, the production of all male tilapia populations can maximize somatic growth. The use of synthetic male steroid, methyltestosterone (MT), in the feeds is the most commercially used method to sexually reverse females. It is at the fry stage that this is administered (Fig. 4), since it must be provided

before, during and after their sexual differentiation and therefore the duration of treatment, fry size, as well as dose are crucial to success (Hiott & Phelps 1993). To provide this treatment prior to the onset of sexual differentiation, the best results are when fry sizes are at ≤ 11 mm for a period of 28 days (Hiott & Phelps 1993) while the most common dose for various tilapia strains is 60 mg MT kg⁻¹ of feed (Tayamen & Shelton 1978; Shelton 1981; Hiott & Phelps 1993). Phelps and Okoko (2011) revealed that the minimum effective dose for Nile tilapia was shown to be 14 mg MT kg⁻¹, and beyond 60 mg MT kg⁻¹ became less effective on sex reversal. It is important to note that dietary methyltestosterone can deteriorate if stored at room temperature (7% reduction in methyltestosterone after 1 month), however, when stored at -20 and -40°C it was still stable after 6 months (Barry *et al.* 2007).

Due to the potential impact to human consumers, radioactively labelled methyltestosterone were measured over time, and from the last feeding, about 90–97% of this hormone was excreted within 24 h while after 21 days it was found at $<1\%$ in both the muscle and viscera (Goudie *et al.* 1986). It was suggested that this would become further diluted with time/growth and when at market size would contain negligible levels. Nevertheless, methyltestosterone is banned in Europe due to concerns about the potential effects to both the consumer as well as the environment and therefore an investigation was made using a natural androgen, 11 β -hydroxyandrostenedione (11 β OHA4), for sex reversing Florida tilapia hybrids (Desprez *et al.* 2003). It was shown that dietary 11 β OHA4 at a concentration of 50 mg kg⁻¹ feed for 28 days successfully produced 99% males and was comparable to methyltestosterone (Desprez *et al.* 2003).

Practical feed formulations: fry to market size

When designing diets for tilapia, the goal is cost-effectively to produce a nutrient dense and highly digestible palatable feed of consistent composition. As previously mentioned, fry are often fed with crumble, however, for fingerlings and throughout the grow-out period, various feed types exist such as sinking pellets, moist pellets and extruded feeds, which can impact tilapia productivity. It was reported in two similar feeding trials on hybrid tilapia (starting weight of 120 g for 119 days and 17 g for 90 days, respectively) that extruded floating pellets led to higher growth than sinking pellets (Creswell 2005). Similarly, Ammar *et al.* (2008) compared extruded and non-extruded (25% crude protein) diets fed to Nile tilapia fingerlings polycultured with grey mullet (*Mugil cephalus*) in earthen ponds and it was found that extruded pellets significantly increased the growth and nutrient utilization of both Nile tilapia and grey mullet.

Generally, to protect their intellectual property, feed companies do not reveal the exact nutrient and ingredient

composition of commercial fish feeds. Nevertheless, least cost tilapia feed formulations as well as vitamin and mineral supplementation have been reported for commercial tilapia feeds used in Asia (Orachunwong *et al.* 2001) (Tables 7 and 8), which can serve as a guide for both farmers and feed manufacturers. Commercial tilapia feeds are commonly grouped as pre-starter, starter, grower or finisher feeds with each feed group sometimes sub-divided further depending on the quality or feed additives used. It is worthwhile noting that not only different protein levels are preset for each life stage, but also the maximum acceptable inclusion level of alternative protein sources concomitant with tilapia development (Table 7). Ingredient use can also vary widely depending on the country with up to 60% of soybean meal and 30% rice bran used in Vietnam as a protein and carbohydrate source, respectively, while in China, protein can be in the form of soybean meal, cottonseed meal and/or rape/canola meal with maize, wheat and grain serving as carbohydrates (Food & Agriculture Organization of the United Nations (FAO) 2011b). While significant amounts of feed in China are made on-farm, often utilizing locally available ingredients that are often produced as moist pellets, these may not be sufficient to meet the nutritional requirements of tilapia or be produced in the most appropriate form. In response to expanding tilapia production, China has permitted the import of both raw materials (e.g. soybean meal) and commercial pellets from different countries at lower tariffs. Moreover, the Chinese government instituted new regulatory rules including inspecting mills for proper hygiene and storage practices as well as testing the feeds for antibiotics and hormones (Food & Agriculture Organization of the United Nations (FAO) 2007). In Vietnam, the Ministry of Fisheries has implemented standards and specifications for formulated tilapia feeds where the composition and physical properties of these feeds are regulated by law to ensure that feed companies comply. This often includes a list of prohibited chemicals, antibiotics, feed ingredients and other substances (Table 9). Some of the prohibited compounds in fish feeds are unique to certain countries, such as the Malaysian Feed Act of 2009 that prohibits the inclusion of porcine products and by-products due to local religious sensitivities. Since most tilapia farming countries have feed regulations that control the manufacture, import and distribution of tilapia feeds, it is hoped that this review will assist policy makers in coming up with and updating the implemented standards and feed specifications.

Genetics and implications to nutrient utilization

With an enhanced understanding and relative ease of genetic selection in tilapia, it is anticipated that genetically improved tilapia strains will continue to be farmed in order

Constraint	% Restriction	Prestarter	Starter	Grower	Finisher
Crude protein	Min	40	30	25	20
Crude fat	Min	4	4	4	4
Crude fibre	Max	4	4	6	8
Lysine	Min	2.04	1.53	1.28	1.02
Methionine + Cystine	Min	1.28	0.96	0.80	0.64
Threonine	Min	1.44	1.08	0.90	0.72
Fatty acid (omega-3)	Min	0.50	0.50	0.50	0.50
Fatty acid (omega-6)	Min	0.50	0.50	0.50	0.50
Calcium	Max	2.50	2.50	2.50	2.50
Phosphorus (total)	Max	1.50	1.50	1.50	1.50
Phosphorus (available)	Min	0.60	0.60	0.60	0.60
Starch	Min	25	25	25	25
Digestible energy (kcal kg ⁻¹)	Min	2800	2800	2800	2800
Vitamin and mineral premix	Fixed	2	2	2	2
Fishmeal (60% protein min)	Min	15	12	10	8
Rice bran			No limit		
Wheat bran			No limit		
Rapeseed/Cottonseed	Max	–	10	15	20
Corn/Cassava/Sorghum	Min	10	10	10	10

Data obtained from Orachunwong *et al.* (2001).

Table 8 Examples of vitamin and mineral supplementation used in commercial tilapia feed

Vitamin	mg kg ⁻¹	IU kg ⁻¹	Mineral	g kg ⁻¹
Thiamine	20		Calcium	3
Pyridoxine	10		Phosphorus	7
Riboflavin	20		Magnesium	0.4
B ₁₂	1		Iron	0.03
Folic acid	5		Zinc	0.2
Choline	500		Copper	0.003
Pantothenic acid	50		Manganese	0.013
Niacin	100		Iodine	0.0025
Ascorbic acid	375			
Biotin	1			
Inositol	100			
K	10			
A		5500		
D ₃		2000		
Alpha-tocopherol		50		

Data obtained from Orachunwong *et al.* (2001).

to increase productivity and thus keep pace with global demands for tilapia products. Currently the 'genetically improved farmed tilapia' (GIFT) and 'genetically male Nile tilapia' (GMNT) are examples of these. The creation of the GIFT strain was undertaken at the WorldFish Center and was selected for the best growth from eight crossbred and pure tilapia strains (four wild and four established strains) (Ponzoni *et al.* 2011) while 'supermale technology' is used to create GMNT by having YY chromosomes and, when

Table 7 Examples of the constraints and restrictions of a least-cost programme for commercial tilapia feed formulation

bred with females (XX), would yield all male populations (XY) (Mair *et al.* 1997). Such tilapia strains with enhanced growth potential have received increasing global attention, however, reports on their productivity and nutrient utilization efficiencies in comparison with conventional tilapia strains have been somewhat contradictory.

Mamun *et al.* (2007a) compared the growth and nutrient digestibilities among GIFT, GMNT and a conventional tilapia strain with each fish housed individually within a recirculating system and found no significant differences, although the GIFT and GMNT tended to have slightly higher growth rates. Moreover, when the metabolic rates were compared, only the active metabolic rate at maintenance conditions for GIFT tilapia were significantly lower than the other fish strains while all other metabolic indicators were similar (Mamun *et al.* 2007b). To reconcile these findings with previous reports of the GIFT and GMNT strains having higher productivity rates under farming conditions (Hussain *et al.* 2000; Ridha 2006; Romana-Eguia *et al.* 2010), Mamun *et al.* (2007b) suggested that these strains would likely eat more as well as being more competitive. However, in an experiment deliberately utilizing fixed feeding rates of 4% BW at two different protein diets of 25% or 35%, it was found that the growth performance, feeding efficiency as well as lipid digestion were significantly higher for the GIFT strain over conventional tilapia red hybrids (Ng & Hanim 2007). Furthermore, the tilapia hybrids fed 25% or 35% protein diets showed no further growth improvement,

Table 9 The quality standard for formulated tilapia feeds as set by the Ministry of Fisheries, Vietnam (28 TCN 189: 2004)

Standard	Feed type based on tilapia weight (g)					
	5	5–10	10–20	20–200	200–500	>500
Pellet diameter (mm)	1.0	1.5	2.0	4.0	4.0	6.0
maximum						
Durability (h)	30	30	30	30	30	30
minimum						
Crude energy (kcal kg ⁻¹ feed) minimum	3200	3000	2860	2800	2750	2700
Crude protein (% weight)	40	35	30	27	25	20
maximum						
Crude lipid (% weight)	6	6	5	5	4	4
Crude fibre (% weight)	5	5	6	6	7	7
maximum						
Ash (% weight)	19	19	19	19	19	19
maximum						
Calcium (% weight)	2.5	2.5	2.5	2.5	2.5	2.5
maximum						
Calcium/phosphorus ratio	1.0–1.5	1.0–1.5	1.0–1.5	1.0–1.5	1.0–1.5	1.0–1.5
NaCl (% weight)	2.5	2.5	2.5	2.5	2.5	2.5
maximum						
Lysine (% weight)	1.7	1.6	1.4	1.3	1.1	0.9
minimum						
Methionine (% weight)	1.0	0.9	0.8	0.7	0.6	0.5
minimum						
Aflatoxin	<10 ppb	<10 ppb	<10 ppb	<10 ppb	<10 ppb	<10 ppb
Antibiotics or chemicals			Not permitted†			

Data modified from Food and Agriculture Organization of the United Nations (FAO) (2007).

†Prohibited at all stages based on Resolution No. 01/2002/QĐ-BTS, 22 January 2002 of Ministry of Fisheries, Vietnam.

while the GIFT strain fed 35% protein diet had a 10% higher final weight than those fed the 25% protein diet, suggesting that appropriate dietary protein levels to support their faster growth rates may be warranted (Ng & Hanim 2007). It should be noted that the better performance of genetically selected strains, including the GIFT strains, versus other Nile tilapia strains have been confirmed in a number of other studies (Hussain *et al.* 2000; Ridha 2006; Romana-Eguia *et al.* 2010). For example, Hussain *et al.* (2000) found that the GIFT strain showed higher growth in various farming conditions (nursery, cisterns, hapas, nets and grow out ponds) and at different sites in Bangladesh, leading to a production of 4410 kg ha⁻¹ for GIFT compared with 2966 kg ha⁻¹ for conventional tilapia strains.

Recently, Teoh *et al.* (2011) evaluated the fatty acid metabolism of the GIFT strain and red hybrid tilapia when fed purified diets with fish oil or vegetable oil blends. It was demonstrated that the fatty acid digestibility was not differ-

ent among the tilapia strains, however, the GIFT strain showed a higher degree of *in vivo* LC-PUFA biosynthesis than the hybrids when fed vegetable oil-based diets and this was reflected in the whole body content (Teoh *et al.* 2011). Moreover, there were indications that the dietary requirements for SFA and MUFA may be higher as an energy source for the GIFT strain. Due to a higher whole body lipid content, it was also suggested that this strain may either have a lower metabolic requirement and/or be more efficient at utilizing carbohydrates (Teoh *et al.* 2011). By analysing over 500 fillets from the GIFT strain over two successive generations, it was found that there was no improvement in the LC-PUFA content compared with controls that were selected for average fecundity (Nguyen *et al.* 2010). Nevertheless, there were indications of the potential to genetically select tilapia to yield a healthier fillet fatty acid profile with higher omega-3 fatty acids, although the estimates of heritability was low (10%) and may be at the expense of other fillet quality traits (Nguyen *et al.* 2010). A

recent study in our laboratory, comparing the growth performance of the GIFT strain with unimproved tilapia strains, indicated that the GIFT strain was more efficient in utilizing SBM-based diets (unpublished results). These initial studies seemed to indicate an enhanced ability of improved tilapia strains to utilize plant-based lipid and protein sources which augurs well for the continued sustainability and scalability of the global tilapia industry. Whether a revision of the established nutrient requirements is needed for these improved strains of farmed tilapia in order for them to express their full genetic potential for growth remains to be investigated.

Finishing diets

In spite of potential genetic selection for increased LC-PUFA synthesis, currently the fatty acid content of tilapia fillets will generally reflect those of the diet. Therefore with increasing use of plant-based oils in commercial tilapia feeds, often characterized by having a high omega-6 and low omega-3 fatty acids content, the resultant tilapia product may be less healthy for the human consumer compared with wild-caught tilapia (Karapanagiotidis *et al.* 2006; Young 2009). In a study examining the fatty acid content among various commonly consumed fish, Weaver *et al.* (2008) found that farmed tilapia had the highest mean n-6/n-3 ratio of over 2.5, an arachidonic acid/eicosapentaenoic acid ratio over 10 as well as high total n-6 levels and suggested that this would be detrimental to the health of human consumers, particularly those predisposed to inflammatory infections. Though somewhat controversial, the findings of Weaver *et al.* (2008) have given farmed tilapia products a poor public image in terms of nutritional value. Modern day human diets are already high in n-6 fatty acids (especially linoleic acid) and we should endeavour to limit the deposition of this fatty acid in farmed tilapia. Moreover, linoleic acid competes with α -linolenic acid for the LC-PUFA synthesis pathway in fish thereby reducing the production of the more physiologically important eicosapentaenoic acid and docosahexaenoic acid (Karapanagiotidis *et al.* 2006; Teoh *et al.* 2011).

Exploiting the strong influence of dietary fatty acids on the final fatty acid profile of tilapia fillets, there has been increasing interest in the use of finishing diets to boost their nutritional value prior to harvest (Ng & Chong 2004; Visentainer *et al.* 2005; Tonial *et al.* 2009; Trushenski *et al.* 2009; Santos *et al.* 2011; Luo *et al.* 2012). As a means to increase n-3 PUFA in tilapia fillets, flaxseed (linseed) oil, which is rich in α -linolenic acid, has been shown to be highly successful, although this depends on the feeding duration (Justi *et al.* 2003; Visentainer *et al.* 2005; Tonial *et al.* 2009). After tilapia were fed with soybean oil-based diets (high in linoleic acid), Tonial *et al.* (2009) found that 45 days was the mini-

mum duration required to reduce the n-6/n-3 fillet ratios from 7.6 to 1.1 when reverted to a flaxseed-based diet. While this had no effect on the eicosapentaenoic acid levels, the docosahexaenoic acid and arachidonic acid levels increased and decreased, respectively (Tonial *et al.* 2009). A similar finding was reported by Visentainer *et al.* (2005) where increasing dietary flaxseed oil, at the expense of sunflower seed oil, significantly increased both the docosahexaenoic acid and eicosapentaenoic acid content of tilapia when sampled after 5 months. It is known that tilapia can synthesize docosahexaenoic acid and eicosapentaenoic acid from α -linolenic acid, and the rate of this synthesis can increase when dietary LC-PUFA and α -linolenic acid decrease and increase, respectively (Olsen *et al.* 1990; Tocher *et al.* 2002; Teoh *et al.* 2011). Nevertheless, since their ability for LC-PUFA synthesis is currently insufficient to equal the fatty acid profile found in fish fed fish oil-based diets (Tocher *et al.* 2002), research has been performed on finishing diets containing only fish oil and/or blends with plant oils with lower n-6 fatty acid levels such as crude palm oil and linseed oil (LO). Trushenski *et al.* (2009) found that tilapia previously fed various alternative lipids in a 1:1 ratio with fish oil for 13 weeks and then fed a fish oil finishing diet for 4 weeks, had LC-PUFA percentages that improved slightly, but not significantly, in each treatment. Ng and Chong (2004) demonstrated that the n-3 PUFA content of tilapia previously fed with crude palm oil or soybean oil-based grow-out diets for 5 months could be significantly improved by switching to a fish oil-based finishing diet for 3 months before harvest. By combining this fish oil finishing diet strategy and using blended oils, tilapia previously fed fish oil + crude palm oil or linseed oil + crude palm oil diets, and switched to a fish oil-based finishing diet for 2 months, had a n-3 PUFA content in fillets not significantly different compared with the fillets of tilapia fed fish oil-based diets throughout the culture cycle. Furthermore, the fillets of tilapia fed diets with plant oils rich in endogenous vitamin E (e.g. CPO and SBO) have enhanced oxidative stability that prolonged their shelf-life in frozen storage (Ng & Chong 2004; Ng & Bahurmiz 2009). There are also likely additional health benefits for the human consumer due to the deposition of tocotrienols in tilapia fillets when fed palm oil-based diets (Wang *et al.* 2006). The role of these minor components of plant oils in enhancing the fillet quality and nutritional value of the final farmed tilapia products warrants more research.

Considering the higher cost of both linseed oil and fish oil as well as the relatively long time frame, it is likely that if tilapia continues to be purchased based on weight rather than their fatty acid content, the use of finishing diets may not be used on a wide scale in tilapia farming practices. This is further compounded by the fact that tilapia are being sold as a 'generic' product, rather than 'branded' which consumers purchase based on quantity, thus present-

ing no motivation or economical justification for farmers to adopt this practice. In the future, this may change, such as the development of niche tilapia markets or through government guidelines on the final product quality.

Conclusions

Throughout this review, it has been shown that nutrition and feeding management can be substantially different at various tilapia culture stages but play an integral role in their productivity. Therefore understanding the nutritional requirements of tilapia throughout their culture cycle and implementing the most appropriate feeding management strategy is important to produce a higher quality product cost-effectively. While the protein and amino acid requirements of tilapia are generally well known and studied, the fatty acid and some vitamin/mineral requirements are less well understood including contradictory results which represent major knowledge gaps that require further investigations. Indeed, perhaps some of contrasting findings among studies may be due to complex interactions among various ingredients or even genetics (use of different tilapia strains) that can influence their utilization. In the case of the latter, this will likely become increasingly important to understand and quantify since there are indications that improved tilapia strains utilize nutrients differently, which may require specific dietary formulations for their full genetic potential to be realized.

It is inevitable that dietary alternatives to marine-based fishmeal and fish oil in tilapia feeds will be used increasingly, thus posing new challenges to tilapia farmers and feed manufacturers around the world. Currently, fishmeal is generally only utilized for the younger tilapia stages and for only a short time before being fed diets with higher alternative-based protein ingredients, particularly soybean meal, during their longer grow-out stages. However, due to the popularity of soybean meal as an alternative to fishmeal in animal feeds, this too is becoming more costly and the current trend is moving towards identifying other suitable alternatives. Research has shown that CSM and R/CM are being increasingly used, however, others such plant protein mixtures and some terrestrial animal proteins can similarly be successfully utilized. The use of other alternatives to fishmeal and/or soybean meal are somewhat country or locality specific. This may be due to a combination of factors, such as not being economically justified due to a limited scale of production/availability or some that are only recently being explored and researched. The use of these sustainable ingredients and at higher inclusion rates may be realized through increasing the understanding of tilapia nutrition, such as methods to mitigate antinutritional factors and appropriate feed additive supplementation. In the next two decades, it is envisaged that more research will be conducted to incor-

porate other plant based ingredients as an alternative to soybean meal in commercial tilapia feeds and with increasing success.

On the other hand, the situation with dietary fish oil alternatives is different since it precludes the possibility of cheaply including dietary omega-3 fatty acids (i.e. DHA and EPA) that are healthy to the consumer. This is perhaps a major constraint and issue with tilapia farming and thus, research should explore ways to select genetically for increased LC-PUFA synthesis in tilapia as well as designing appropriate cost-effective finishing diets to boost their nutritional benefits. Nevertheless, these concerns do not apply to broodstock where dietary crude palm oil and soybean oil have been shown to enhance the reproductive performance of tilapia over fish oil-based diets. Moreover, appropriate feeding regimes can be adopted to help synchronize fry production, which is essential to ensure uninterrupted tilapia production on a large scale. Due to the worldwide popularity of tilapia and their continual high demand, it is anticipated that nutritional research will continue. Much focused research is still required on the impact of nutrition on tilapia health, gene–nutrient interactions and post-harvest quality, as well as its impact on the environment. Through concerted efforts, tilapia farming will continue to expand and thrive in an environmentally friendly and sustainable way, while meeting the increasing seafood demand from a growing human population.

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References

- Abdelhamid AM, Mehri AI, El-Barbary MI, El-Sharawy MA (2010) An attempt to improve the reproductive efficiency of Nile tilapia brood stock fish. *Fish Physiology and Biochemistry* **36**: 1097–1104.
- Abdel-Tawwab M, Ahmad MH, Khattab YAE, Shalaby AME (2010) Effect of dietary protein level, initial body weight, and their interaction on the growth, feed utilization, and physiological alterations of Nile tilapia, *Oreochromis niloticus* (L.). *Aquaculture* **298**: 267–274.
- Akinleye AO, Kumar V, Makkar HPS, Angulo-Escalante MA, Becker K (2012) *Jatropha platyphylla* kernel meal as feed ingredient for Nile tilapia (*Oreochromis niloticus* L.): growth, nutrient utilization and blood parameters. *Journal of Animal Physiology and Animal Nutrition* **96**: 119–129.

- Al-Hafedh YS, Siddiqui AQ (1998) Evaluation of guar seed as a protein source in Nile tilapia, *Oreochromis niloticus* (L.), practical diets. *Aquaculture Research* **29**: 703–708.
- Amirkolaie AK, Verreth JAJ, Schrama JW (2006) Effect of gelatinization degree and inclusion level of dietary starch on the characteristics of digesta and faeces in Nile tilapia (*Oreochromis niloticus* (L.)). *Aquaculture* **260**: 194–205.
- Ammar AA, Abd-Elgawad AS, Salama AA (2008) Effect of extruded and non-extruded fish pellet on growth performance and total production of Nile tilapia and grey mullet fingerlings reared in a polyculture system in earthen ponds. In: 8th International Symposium on Tilapia in Aquaculture, Cairo, Egypt, October 12–14, pp. 823–834.
- Anderson J, Jackson AJ, Matty AJ, Capper BS (1984) Effects of dietary carbohydrate and fibre on the tilapia *Oreochromis niloticus* (Linn.). *Aquaculture* **37**: 303–314.
- Ayyat MS, El-Marakby HI, Sharaf SM (2011) Effect of dietary protein level, stocking density, and dietary pantothenic acid supplementation rate on performance and blood components of Nile tilapia *Oreochromis niloticus*. *Journal of Applied Aquaculture* **23**: 122–135.
- Bahurmiz OM, Ng WK (2007) Effects of dietary palm oil source on growth, tissue fatty acid composition and nutrient digestibility of red hybrid tilapia, *Oreochromis* sp., raised from stocking to marketable size. *Aquaculture* **262**: 382–392.
- Barros MM, Ranzani-Paiva MJT, Pezzato LE, Falcon DR, Guimarães IG (2009) Haematological response and growth performance of Nile tilapia (*Oreochromis niloticus* L.) fed diets containing folic acid. *Aquaculture Research* **40**: 895–903.
- Barry TP, Marwah A, Marwah P (2007) Stability of 17 α -methyl-testosterone in fish feed. *Aquaculture* **271**: 523–529.
- Belal IEH (2008) Evaluating fungi-degraded date pits as a feed ingredient for Nile tilapia *Oreochromis niloticus* L. *Aquaculture Nutrition* **14**: 445–452.
- Bhujel RC (2000) A review of strategies for the management of Nile tilapia (*Oreochromis niloticus*) broodfish in seed production systems, especially hapa-based systems. *Aquaculture* **181**: 37–59.
- Bhujel RC, Yakupitiyage A, Turner WA, Little DC (2001) Selection of a commercial feed for Nile tilapia (*Oreochromis niloticus*) broodfish breeding in a hapa-in-pond system. *Aquaculture* **194**: 303–314.
- Bhujel RC, Little DC, Hossain A (2007) Reproductive performance and the growth of pre-stunted and normal Nile tilapia (*Oreochromis niloticus*) broodfish at varying feeding rates. *Aquaculture* **273**: 71–79.
- Bijvelds MJC, Flik G, Bonga SEW (1997) Mineral balance in *Oreochromis mossambicus*: dependence on magnesium in diet and water. *Fish Physiology and Biochemistry* **16**: 323–331.
- Bishop CD, Angus RA, Watts SA (1995) The use of feather meal as a replacement for fish meal in the diet of *Oreochromis niloticus* fry. *Bioresource Technology* **54**: 291–295.
- Borgeson TL, Racz VJ, Wilkie DC, White LJ, Drew MD (2006) Effect of replacing fishmeal and oil with simple or complex mixtures of vegetable ingredients in diets fed to Nile tilapia (*Oreochromis niloticus*). *Aquaculture Nutrition* **12**: 141–149.
- Brett JR (1973) Energy expenditure of sockeye salmon *Oncorhynchus nerka*, during sustained performance. *Journal of the Fisheries Research Board of Canada* **30**: 1799–1809.
- do Carmo e Sá MV, Pezzato LD, Lima MMBF, Padilha PM (2004) Optimum zinc supplementation level in Nile tilapia *Oreochromis niloticus* juveniles diets. *Aquaculture* **238**: 385–401.
- Chen W, Ai Q, Mai K, Xu W, Liufu Z, Zhang W *et al.* (2011) Effects of dietary soybean saponins on feed intake, growth performance, digestibility and intestinal structure in juvenile Japanese flounder (*Paralichthys olivaceus*). *Aquaculture* **318**: 95–100.
- Cheng ZJ, Hardy RW (2002) Apparent digestibility coefficients and nutritional value of cottonseed meal for rainbow trout (*Oncorhynchus mykiss*). *Aquaculture* **212**: 361–372.
- Chou BS, Shiau SY (1996) Optimal dietary lipid level for growth of juvenile hybrid tilapia, *Oreochromis niloticus* $\times$ *Oreochromis aureus*. *Aquaculture* **143**: 185–195.
- Chou BS, Shiau SY, Hung SSO (2001) Effect of dietary cod liver oil on growth and fatty acids of juvenile hybrid tilapia. *North American Journal of Aquaculture* **63**: 277–284.
- Cnaani A, Barki A, Slosman T, Scharcanski A, Milstein A, Harpaz S (2010) Dietary salt supplement increases the growth rate in freshwater cultured tilapia hybrids. *Aquaculture Research* **41**: 1545–1548.
- Coyle SD, Mengel GJ, Tidwell JH, Webster CD (2004) Evaluation of growth, feed utilization, and economics of hybrid tilapia, *Oreochromis niloticus* $\times$ *Oreochromis aureus*, fed diets containing different protein sources in combination with distillers dried grains with solubles. *Aquaculture Research* **35**: 365–370.
- Creswell D (2005) The feeding and nutrition of the tilapia. Part 2. Feed management. *AQUA Culture AsiaPacific Magazine November/December*: 32–33.
- Dabrowska H, Meyer-Burgdorff K, Günther KD (1989) Interaction between dietary protein and magnesium level in tilapia (*Oreochromis niloticus*). *Aquaculture* **76**: 277–291.
- Davies SJ, McConnell S, Bateson RI (1990) Potential of rapeseed meal as an alternative protein source in complete diets for tilapia (*Oreochromis mossambicus* Peters). *Aquaculture* **87**: 145–154.
- De Silva SS (1985) Performance of *Oreochromis niloticus* (L.) fry maintained on mixed feeding schedules of different dietary protein. *Aquaculture and Fisheries Management* **16**: 331–340.
- De Silva SS, Perera MK (1984) Digestibility in *Sarotherodon niloticus* fry: effect of dietary protein level and salinity with further observations on variability in daily digestibility. *Aquaculture* **38**: 293–306.
- De Silva SS, Gunasekera RM, Atapattu D (1989) The dietary protein requirements of young tilapia and an evaluation of the least cost dietary protein levels. *Aquaculture* **80**: 271–284.
- De Silva SS, Gunasekera RM, Shim KF (1991) Interactions of varying dietary protein and lipid levels in young red tilapia: evidence of protein sparing. *Aquaculture* **95**: 305–318.

- Desprez D, G  raz E, Hoareau MC, M  lard C, Bosc P, Baroiller JF (2003) Production of a high percentage of male offspring with a natural androgen 11 β -hydroxyandrostenedione (11 β OHAA4), in Florida red tilapia. *Aquaculture* **216**: 55–65.
- Eid AE, Ghonim SI (1994) Dietary zinc requirement of fingerling *Oreochromis niloticus*. *Aquaculture* **119**: 259–264.
- Eknath AE, Hulata G (2009) Use and exchange of genetic resources of Nile tilapia (*Oreochromis niloticus*). *Reviews in Aquaculture* **1**: 197–213.
- El-Dahhar AA, Lovell RT (1995) Effect of protein to energy ratio in purified diets on growth performance, feed utilization and body composition of Mozambique tilapia, *Oreochromis mossambicus* (Peters). *Aquaculture Research* **26**: 451–457.
- Eleraky W, Saleh G, Gropp JM (1995) A short note on the vitamin E requirement of *Tilapia nilotica* (*Oreochromis niloticus*) – effects on health and growth. *Journal of Applied Ichthyology* **11**: 375–377.
- El-Husseiny OM, El Din G, Abdul-Aziz M, Mabroke RS (2008) Effect of mixed protein schedules combined with choline and betaine on the growth performance of Nile tilapia (*Oreochromis niloticus*). *Aquaculture Research* **39**: 291–300.
- El-Saidy DMSD (2011) Effect of using okara meal, a by-product from soymilk production as a dietary protein source for Nile tilapia (*Oreochromis niloticus* L.) mono-sex males. *Aquaculture Nutrition* **17**: 380–386.
- El-Saidy DMSD, Gaber MMA (2003) Replacement of fish meal with a mixture of different plant protein sources in juvenile Nile tilapia, *Oreochromis niloticus* (L.) diets. *Aquaculture Research* **34**: 1119–1127.
- El-Saidy DMSD, Gaber MM (2004) Use of cottonseed meal supplemented with iron for detoxification of gossypol as a total replacement of fish meal in Nile tilapia, *Oreochromis niloticus* (L.) diets. *Aquaculture Research* **35**: 859–865.
- El-Saidy DMSD, Saad AS (2011) Effects of partial and complete replacement of soybean meal with cottonseed meal on growth, feed utilization and haematological indexes for mono-sex male Nile tilapia, *Oreochromis niloticus* (L.) fingerlings. *Aquaculture Research* **42**: 351–359.
- El-Sayed AFM (1990) Long-term evaluation of cotton seed meal as a protein source for Nile tilapia, *Oreochromis niloticus* (Linn.). *Aquaculture* **84**: 315–320.
- El-Sayed AFM (1998) Total replacement of fish meal with animal protein sources in Nile tilapia, *Oreochromis niloticus* (L.), feeds. *Aquaculture Research* **29**: 275–280.
- El-Sayed AFM (1999) Alternative dietary protein sources for farmed tilapia, *Oreochromis* spp. *Aquaculture* **179**: 149–168.
- El-Sayed AFM (2002) Effects of stocking density and feeding levels on growth and feeding efficiency of Nile tilapia (*Oreochromis niloticus* L.) fry. *Aquaculture Research* **33**: 621–626.
- El-Sayed AFM (2006) *Tilapia Culture*. CABI Publishing, Oxfordshire, UK.
- El-Sayed AFM, Garling DL (1988) Carbohydrate-to-lipid ratios in diets for *Tilapia zillii* fingerlings. *Aquaculture* **73**: 157–163.
- El-Sayed AFM, Kawanna M (2008) Effects of dietary protein and energy levels on spawning performance of Nile tilapia (*Oreochromis niloticus*) broodstock in a recycling system. *Aquaculture* **280**: 179–184.
- El-Sayed AFM, Mart  nez IN, Moyano FJ (2000) Assessment of the effect of plant inhibitors on digestive proteases of Nile tilapia using *in vitro* assays. *Aquaculture International* **8**: 403–415.
- El-Sayed AFM, Mansour CR, Ezzat AA (2003) Effects of dietary protein level on spawning performance of Nile tilapia (*Oreochromis niloticus*) broodstock reared at different water salinities. *Aquaculture* **220**: 619–632.
- El-Sayed AFM, Mansour CR, Ezzat AA (2005) Effects of dietary lipid source on spawning performance of Nile tilapia (*Oreochromis niloticus*) broodstock reared at different salinities. *Aquaculture* **248**: 187–196.
- Fasakin EA, Serwata RD, Davies SJ (2005) Comparative utilization of rendered animal derived products with or without composited mixture of soybean meal in hybrid tilapia (*Oreochromis niloticus* $\times$ *Oreochromis mossambicus*) diets. *Aquaculture* **249**: 329–338.
- Fitzsimmons K, Dickenson G, Brand C, Davis J (1997) Effects of reducing dietary lipid levels on growth and body composition of hybrid tilapia in an intensive recirculating-water system. *Progressive Fish Culturist* **59**: 293–296.
- Food and Agriculture Organization of the United Nations (FAO) (2007). Study and analysis of feeds and fertilizers for sustainable aquaculture development. In: Hasan MR, Hecht T, De Silva SS, Tacon AGJ (eds) *FAO Fisheries and Technical Paper* 497. pp. 1–510. Food and Agriculture Organization of the United Nations, Rome.
- Food and Agriculture Organization of the United Nations (FAO) (2011a) *FISHSTAT plus: Universal Software for Fishery Statistical Time Series, Version 2.3*. FAO Fisheries Department, Fishery Information, Data and Statistical Unit, Rome.
- Food and Agriculture Organization of the United Nations (FAO) (2011b) Demand and Supply of Feed Ingredients for Farmed Fish and Crustaceans – Trends and Prospects. In: Tacon AGJ, Hasan MR, Metian M (eds) *FAO Fisheries and Aquaculture Technical Paper* 564. Food and Agriculture Organization of the United Nations, Rome.
- Food and Agriculture Organization of the United Nations (FAO) (2011c) *FAOSTAT: FAO Food and Agricultural Commodities Production*. FAO, Rome.
- Fowler LG (1990) Feather meal as a dietary protein source during parr-smolt transformation in fall chinook salmon. *Aquaculture* **89**: 301–314.
- Francis G, Makkar HPS, Becker K (2001) Antinutritional factors present in plant-derived alternate fish feed ingredients and their effects in fish. A review. *Aquaculture* **199**: 197–227.
- Furuya WM, Pezzato LE, Barros MM, Pezzato AC, Furuya VRB, Miranda EC (2004) Use of ideal protein concept for precision formulation of amino acid levels in fish-meal-free diets for juvenile Nile tilapia (*Oreochromis niloticus* L.). *Aquaculture Research* **35**: 1110–1116.
- Gaber MMA (1996) Effect of oil source on growth, feed conversion and body composition of Nile tilapia (*Oreochromis niloticus*). *Annals of Agricultural Science* **34**: 1549–1562.

- Gao W, Liu YJ, Tian LX, Mai KS, Liang GY, Yang HJ *et al.* (2011) Protein-sparing capability of dietary lipid in herbivorous and omnivorous freshwater finfish: a comparative case study on grass carp (*Ctenopharyngodon idella*) and tilapia (*Oreochromis niloticus* × *O. aureus*). *Aquaculture Nutrition* **17**: 2–12.
- Garcia-Abiado MA, Mbahinzireki G, Rinchar J, Lee KJ, Dabrowski K (2004) Effect of diets containing gossypol on blood parameters and spleen structure in tilapia, *Oreochromis* sp., reared in a recirculating system. *Journal of Fish Diseases* **27**: 359–368.
- Gaye-Siessegger J, Focken U, Abel HJ, Becker K (2007) Influence of dietary non-essential amino acid profile on growth performance and amino acid metabolism of Nile tilapia, *Oreochromis niloticus* (L.). *Comparative Biochemistry and Physiology* **146A**: 71–77.
- Getinet GT (2008) Effects of maternal age on fecundity, spawning interval, and egg quality of Nile tilapia, *Oreochromis niloticus* (L.). *Journal of the World Aquaculture Society* **39**: 671–677.
- Goda AM, Wafa ME, El-Haroun ER, Chowdhury MAK (2007) Growth performance and feed utilization of Nile tilapia *Oreochromis niloticus* (Linnaeus, 1758) and tilapia galilae *Sarotherodon galilaeus* (Linnaeus, 1758) fingerlings fed plant protein-based diets. *Aquaculture Research* **38**: 827–837.
- Goddard S, Al-Shagaa G, Ali A (2008) Fisheries by-catch and processing waste meals as ingredients in diets for Nile tilapia, *Oreochromis niloticus*. *Aquaculture Research* **39**: 518–525.
- Goudie CA, Shelton WL, Parker NC (1986) Tissue distribution and elimination of radiolabelled methyltestosterone fed to sexually undifferentiated blue tilapia. *Aquaculture* **58**: 215–226.
- Guimarães IG, Pezzato LE, Barros MM (2008) Amino acid availability and protein digestibility of several protein sources for Nile tilapia, *Oreochromis niloticus*. *Aquaculture Nutrition* **14**: 396–404.
- Gunasekera RM, Shim KF, Lam TJ (1995) Effect of dietary protein level on puberty, oocyte growth and egg chemical composition in tilapia, *Oreochromis niloticus* (L.). *Aquaculture* **134**: 169–183.
- Gunasekera RM, Shim KF, Lam TJ (1996) Influence of protein content of broodstock diets on larval quality and performance in Nile tilapia, *Oreochromis niloticus* (L.). *Aquaculture* **146**: 245–259.
- Gunasekera RM, Shim KF, Lam TJ (1997) Influence of dietary protein content on the distribution of amino acids in oocytes, serum and muscle of Nile tilapia, *Oreochromis niloticus* (L.). *Aquaculture* **152**: 205–221.
- Guo R, Lim C, Xia H, Aksor MY, Li M (2010) Effect of various dietary vitamin A levels on growth performance and immune response of tilapia (*Oreochromis niloticus*). *Frontiers of Agriculture in China* **4**: 507–512.
- Hajizadeh A, Jauncey K, Rana K (2008) Effects of the dietary lipid source on egg and larval quality of Nile tilapia, *Oreochromis niloticus* (L.). *8th International Symposium on Tilapia in Aquaculture* **2**: 965–977.
- Hamre K (2011) Metabolism, interactions, requirements and functions of vitamin E in fish. *Aquaculture Nutrition* **17**: 98–115.
- Hajzadeh A, Jauncey K, Rana K (2008) Effects of dietary lipid source on egg and larval quality of Nile tilapia, *Oreochromis niloticus* (L.). In: Elghobashy H, Fitzsimmons K, Diab AS (eds) *The Eighth International Symposium on Tilapia in Aquaculture* vol. 2, pp. 965–977; 12–14 Oct 2008, Cairo, Egypt, American Tilapia Association, Pine Bluff.
- Hernández C, Olvera-Novoa MA, Hardy RW, Hermosillo A, Reyes C, González B (2010) Complete replacement of fish meal by porcine and poultry by-product meals in practical diets for fingerling Nile tilapia *Oreochromis niloticus*: digestibility and growth performance. *Aquaculture Nutrition* **16**: 44–53.
- Hiott AE, Phelps RP (1993) Effects of initial age and size on sex reversal of *Oreochromis niloticus* fry using methyltestosterone. *Aquaculture* **112**: 301–308.
- Hossain MA, Focken U, Becker K (2003) Antinutritive effects of galactomannan-rich endosperm of Sesbania (*Sesbania aculeata*) seeds on growth and feed utilization in tilapia, *Oreochromis niloticus*. *Aquaculture Research* **34**: 1171–1179.
- Hsieh SL, Shiau SY (2000) Effects of diets containing different carbohydrates on starved condition in juvenile tilapia *Oreochromis niloticus* × *O. aureus*. *Fisheries Science* **66**: 32–37.
- Hu CJ, Chen SM, Pan CH, Huang CH (2006) Effects of dietary vitamin A or β -carotene concentrations on growth of juvenile hybrid tilapia, *Oreochromis niloticus* × *O. aureus*. *Aquaculture* **253**: 602–607.
- Huang CH, Huang SL (2004) Effect of dietary vitamin E on growth, tissue lipid peroxidation, and liver glutathione level of juvenile hybrid tilapia, *Oreochromis niloticus* × *O. aureus*, fed oxidized oil. *Aquaculture* **237**: 381–389.
- Huang CH, Huang MC, Hou PC (1998) Effect of dietary lipids on fatty acid composition and lipid peroxidation in sarcoplasmic reticulum of hybrid tilapia, *Oreochromis niloticus* × *O. aureus*. *Comparative Biochemistry and Physiology* **120B**: 331–336.
- Huang CH, Chang RJ, Huang SL, Chen W (2003) Dietary vitamin E supplementation affects tissue lipid peroxidation of hybrid tilapia, *Oreochromis niloticus* × *O. aureus*. *Comparative Biochemistry and Physiology* **134A**: 265–270.
- Hussain MG, Kohinoor AHM, Islam MS, Mahata SC, Ali MZ, Tanu MB *et al.* (2000) Genetic evaluation of GIFT and existing strains of Nile tilapia, *Oreochromis niloticus* L., under on-station and on-farm conditions in Bangladesh. *Asian Fisheries Science* **13**: 117–126.
- Ibrahim MD, Fathi M, Mesalhy S, El-Aty AMA (2010) Effect of dietary supplementation of inulin and vitamin C on the growth, hematology, innate immunity, and resistance of Nile tilapia (*Oreochromis niloticus*). *Fish and Shellfish Immunology* **29**: 241–246.
- Izquierdo MS, Fernandez-Palacios H, Tacon AGJ (2001) Effect of broodstock nutrition on reproductive performance of fish. *Aquaculture* **197**: 25–42.

- Jackson AJ, Capper BS, Matty AJ (1982) Evaluation of some plant proteins in complete diets for the tilapia *Sarotherodon mossambicus*. *Aquaculture* **27**: 97–109.
- Jauncey K (2000). Nutritional requirements. In: Beveridge MCM, McAndrew BJ (eds) *Tilapias: Biology and Exploitation*, pp. 327–375. Kluwer Academic Publishers, Dordrecht.
- Justi KC, Hayashi C, Visentainer JV, de Souza NE, Matsushita M (2003) The influence of feed supply time on the fatty acid profile of Nile tilapia (*Oreochromis niloticus*) fed on a diet enriched with n-3 fatty acids. *Food Chemistry* **80**: 489–493.
- Kamat JP, Sarma HD, Devasagayam TPA, Nesaretnam K, Basiron Y (1997) Tocotrienols from palm oil as effective inhibitors of protein oxidation and lipid peroxidation in rat liver microsomes. *Molecular and Cell Biochemistry* **170**: 131–138.
- Kanazawa A, Teshima SI, Sakamoto M, Awal MA (1980) Requirement of *Tilapia zillii* for essential fatty acids. *Bulletin of the Japanese Society of Scientific Fisheries* **46**: 1353–1356.
- Karapanagiotidis IT, Bell MV, Little DC, Yakupitiyage A, Rakshit SK (2006) Polyunsaturated fatty acid content of wild and farmed tilapias in Thailand: effect of aquaculture practices and implications for human nutrition. *Journal of Agricultural and Food Chemistry* **54**: 4304–4310.
- Kasper CS, White MR, Brown PB (2000) Choline is required by tilapia when methionine is not in excess. *Journal of Nutrition* **130**: 238–242.
- Kasper CS, White MR, Brown PB (2002) Betaine can replace choline in diets for juvenile Nile tilapia, *Oreochromis niloticus*. *Aquaculture* **205**: 119–126.
- Keembiyehetty CN, De Silva SS (1993) Performance of juvenile *Oreochromis niloticus* (L.) reared on diets containing cowpea, *Vigna catianga*, and black gram, *Phaseolus mungo*, seeds. *Aquaculture* **112**: 207–215.
- Kihara M, Sakata T (1997) Fermentation of dietary carbohydrates to short-chain fatty acids by gut microbes and its influence on intestinal morphology of a detritivorous teleost tilapia (*Oreochromis niloticus*). *Comparative Biochemistry and Physiology* **118A**: 1201–1207.
- Kim KW, Wang X, Choi SM, Park GJ, Koo JW, Cai SC (2003) No synergistic effects by the dietary supplementation of ascorbic acid, α -tocopheryl acetate and selenium on the growth performance and challenge test of *Edwardsiella tarda* in fingerling Nile tilapia, *Oreochromis niloticus* L. *Aquaculture Research* **34**: 1053–1058.
- Kocher A, Choct M, Porter MD, Broz J (2002) Effects of feed enzymes on nutritive value of soyabean meal fed to broilers. *British Poultry Science* **43**: 54–63.
- Krossøy C, Waagbø R, Ørnsrud R (2011) Vitamin K in fish nutrition. *Aquaculture Nutrition* **17**: 585–594.
- Larumbe-Morán E, Hernández-Vergara MP, Olvera-Novoa MA, Rostro CIP (2010) Protein requirements of Nile tilapia (*Oreochromis niloticus*) fry cultured at different salinities. *Aquaculture Research* **41**: 1150–1157.
- Lavorgna MW, Hafez Y, Handwerker T, Hughes SG (2003) Phosphorus digestibility and activity of intestinal phytase in hybrid tilapia, *Oreochromis niloticus* $\times$ *O. aureus*. *Journal of Applied Aquaculture* **14**: 89–100.
- Leenhouwers JI, Ortega RC, Verreth JAJ, Schrama JW (2007) Digesta characteristics in relation to nutrient digestibility and mineral absorption in Nile tilapia (*Oreochromis niloticus* L.) fed cereal grains of increasing viscosity. *Aquaculture* **273**: 556–565.
- Leenhouwers JI, Pellikaan WF, Huizing HFA, Coolen ROM, Verreth JAJ, Schrama JW (2008) Fermentability of carbohydrates in an *in vitro* batch culture method using inocula from Nile tilapia (*Oreochromis niloticus*) and European sea bass (*Dicentrarchus labrax*). *Aquaculture Nutrition* **14**: 523–532.
- Li JS, Li JL, Wu TT (2007) The effects of copper, iron and zinc on digestive enzyme activity in the hybrid tilapia *Oreochromis niloticus* (L.) $\times$ *Oreochromis aureus* (Steindachner). *Journal of Fish Biology* **71**: 1788–1798.
- Li JS, Li JL, Wu TT (2009) Effects of non-starch polysaccharides enzyme, phytase and citric acid on activities of endogenous digestive enzymes of tilapia (*Oreochromis niloticus* $\times$ *Oreochromis aureus*). *Aquaculture Nutrition* **15**: 415–420.
- Liebert F, Portz L (2005) Nutrient utilization of Nile tilapia *Oreochromis niloticus* fed plant based low phosphorus diets supplemented with graded levels of different sources of microbial phytase. *Aquaculture* **248**: 111–119.
- Lim C, Lemaster B, Brock JA (1993) Riboflavin requirement of fingerling red hybrid tilapia grown in seawater. *Journal of the World Aquaculture Society* **24**: 451–458.
- Lim HA, Ng WK, Lim SL, Ibrahim CO (2001) Contamination of palm kernel meal with *Aspergillus flavus* affects its nutritive value in pelleted feed for tilapia, *Oreochromis mossambicus*. *Aquaculture Research* **32**: 895–905.
- Lim C, Yildirim-Aksoy M, Li MH, Welker TL, Klesius PH (2009) Influence of dietary levels of lipid and vitamin E on growth and resistance of Nile tilapia to *Streptococcus iniae* challenge. *Aquaculture* **298**: 76–82.
- Lim C, Yildirim-Aksoy M, Barros MM, Klesius P (2011a) Thiamin requirement of Nile tilapia, *Oreochromis niloticus*. *Journal of the World Aquaculture Society* **42**: 824–833.
- Lim C, Yildirim-Aksoy M, Klesius P (2011b) Lipid and fatty acid requirements of tilapias. *North American Journal of Aquaculture* **73**: 188–193.
- Lin SC, Liou CH, Shiua SY (2000) Renal threshold for urinary glucose excretion by tilapia in response to orally administered carbohydrates and injected glucose. *Fish Physiology and Biochemistry* **23**: 127–132.
- Liti DM, Waidbacher H, Straif M, Mbaluka RK, Munguti JM, Kyenze MM (2006) Effects of partial and complete replacement of freshwater shrimp meal (*Caridina niloticus* Roux) with a mixture of plant protein sources on growth performance of Nile tilapia (*Oreochromis niloticus* L.) in fertilized ponds. *Aquaculture Research* **37**: 477–483.
- Lovell RT, Limsuwan T (1982) Intestinal synthesis and dietary nonessentiality of vitamin B12 for *Tilapia nilotica*. *Transactions of the American Fisheries Society* **111**: 485–490.
- Luo Z, Tan XY, Liu XJ, Wen H (2011) Effect of dietary betaine levels on growth performance and hepatic intermediary

- metabolism of GIFT strain of Nile tilapia *Oreochromis niloticus* reared in freshwater. *Aquaculture Nutrition* **17**: 361–367.
- Luo Z, Tan XY, Liu CX, Li XD, Liu XJ, Xi WQ (2012) Effect of dietary conjugated linoleic acid levels on growth performance, muscle fatty acid profile, hepatic intermediary metabolism and antioxidant responses in genetically improved farmed Tilapia strain of Nile tilapia *Oreochromis niloticus*. *Aquaculture Research* **43**: 1392–1403.
- Lupatsch I, Deshev R, Magen I (2010) Energy and protein demands for optimal egg production including maintenance requirements of female tilapia *Oreochromis niloticus*. *Aquaculture Research* **41**: 763–769.
- Lusas EW, Riaz MN (1995) Soy protein products: processing and use. *Journal of Nutrition* **125**: 573S–580S.
- MacIntosh DJ, De Silva SS (1984) The influence of stocking density and food ration on fry survival and growth in *Oreochromis mossambicus* and *O. niloticus* female $\times$ *O. aureus* male hybrids reared in a closed circulated system. *Aquaculture* **41**: 345–358.
- Mair GC, Abucay JS, Skibinski DOF, Abella TA, Beardmore JA (1997) Genetic manipulation of sex ratio for the large-scale production of all-male tilapia, *Oreochromis niloticus*. *Canadian Journal of Fisheries and Aquatic Science* **54**: 396–404.
- Mamun SM, Focken U, Becker K (2007a) Comparative digestion efficiencies in conventional, genetically improved and genetically male Nile tilapia, *Oreochromis niloticus* (L.). *Aquaculture Research* **38**: 381–387.
- Mamun SM, Focken U, Becker K (2007b) Comparison of metabolic rates and feed nutrient digestibility in conventional, genetically improved (GIFT) and genetically male (GMNT) Nile tilapia, *Oreochromis niloticus* (L.). *Comparative Biochemistry and Physiology*. **148A**: 214–222.
- Marchetti M, Tossani N, Marchetti S, Baucé G (1999) Stability of crystalline and coated vitamins during manufacture and storage of fish feeds. *Aquaculture Nutrition* **5**: 115–120.
- Matsiko DS, Rurangwa E, Charo-Karisa H, Fred M (2010) Effect of feeding level of Ugachick feed on the reproductive performance and growth of Nile tilapia (*Oreochromis niloticus*) in ponds in Uganda. *Aquaculture Research* **41**: 949–951.
- Mbahinzireki GB, Dabrowski K, Lee KJ, El-Saidy D, Wisner ER (2001) Growth, feed utilization and body composition of tilapia (*Oreochromis* sp.) fed with cottonseed meal-based diets in a recirculating system. *Aquaculture Nutrition* **7**: 189–200.
- Mironova NV (1977) Energy expenditure on egg production in young *Tilapia mossambica* and the influence of maintenance conditions on their reproductive intensity. *Journal of Ichthyology* **17**: 627–633.
- National Research Council (NRC) (2011) *Nutrient Requirements of Fish and Shrimp*. National Academy Press, Washington DC.
- Ng WK, Bahurmiz OM (2009) The impact of dietary oil source and frozen storage on the physical, chemical and sensorial quality of fillets from market-size red hybrid tilapia, *Oreochromis* sp. *Food Chemistry* **113**: 1041–1048.
- Ng WK, Chong CY (2004) An overview of lipid nutrition with emphasis on alternative lipid sources in tilapia feeds. In: 6th International Symposium on Tilapia Aquaculture. Manila, Philippines, Sept. 12–16, pp. 241–218.
- Ng WK, Gibon V (2011) Palm oil and saturated fatty acid-rich vegetable oils. In: Turchini GM, Ng WK, Tocher DR (eds) *Fish Oil Replacement and Alternative Lipid Sources in Aquaculture Feeds*, pp. 99–132. CRC Press, UK.
- Ng WK, Hanim R (2007) Performance of genetically improved Nile tilapia compared with red hybrid tilapia fed diets containing two protein levels. *Aquaculture Research* **38**: 965–972.
- Ng WK, Wang Y (2011) Inclusion of crude palm oil in the broodstock diets of female Nile tilapia, *Oreochromis niloticus*, resulted in enhanced reproductive performance compared to broodfish fed diets with added fish oil or linseed oil. *Aquaculture* **314**: 122–131.
- Ng WK, Lim PK, Sidek H (2001) The influence of a dietary lipid source on growth, muscle fatty acid composition and erythrocyte osmotic fragility of hybrid tilapia. *Fish Physiology and Biochemistry* **25**: 301–310.
- Ng WK, Lim HA, Lim SL, Ibrahim CO (2002) Nutritive value of palm kernel meal pretreated with enzyme or fermented with *Trichoderma koningii* (Oudemans) as a dietary ingredient for red hybrid tilapia (*Oreochromis* sp.). *Aquaculture Research*. **33**: 1199–1207.
- Ng WK, Koh CB, Din ZB (2006) Palm oil-laden spent bleaching clay as a substitute for marine fish oil in the diets of Nile tilapia, *Oreochromis niloticus*. *Aquaculture Nutrition* **12**: 459–468.
- Ng WK, Teh SW, Chowdhury KMA, Bureau DP (2011) Commercial, farm-made and supplementary feeds used by tilapia farmers in Malaysia. *Practical Asian Aquaculture* **2**: 16–20.
- Ngamsnae P, De Silva SS, Gunasekera RM (1999) Arginine and phenylalanine requirement of juvenile silver perch, *Bidyanus bidyanus* and validation of the use of body amino acid composition for estimating individual amino acid requirements. *Aquaculture Nutrition* **5**: 173–180.
- Nguyen TN, Davis DA (2009) Methionine requirement in practical diets of juvenile Nile tilapia, *Oreochromis niloticus*. *Journal of the World Aquaculture Society* **40**: 410–416.
- Nguyen NH, Ponzoni RW, Yee HY, Abu-Bakar KR, Hamzah A, Khaw HL (2010) Quantitative genetic basis of fatty acid composition in the GIFT strain of Nile tilapia (*Oreochromis niloticus*) selected for high growth. *Aquaculture* **309**: 66–74.
- O'Connell JP, Gatlin DM (1994) Effects of dietary calcium and vitamin D3 on weight gain and mineral composition of the blue tilapia (*Oreochromis aureus*) in low-calcium water. *Aquaculture* **125**: 107–117.
- Olivera-Castillo L, Pino-Aguilar M, Lara-Flores M, Granados-Puerto S, Montero-Muñoz J, Olvera-Novoa MA et al. (2011) Substitution of fish meal with raw or treated cowpea (*Vigna unguiculata* L Walp, IT86-D710) meal in diets for Nile tilapia (*Oreochromis niloticus* L.) fry. *Aquaculture Nutrition* **17**: e101–e111.
- Olsen RE, Henderson RJ, McAndrew BJ (1990) The conversion of linoleic acid and linolenic acid to longer chain polyunsaturated fatty acids by tilapia (*Oreochromis niloticus*) in vivo. *Fish Physiology and Biochemistry* **8**: 261–270.

- Olvera-Novoa MA, Olivera-Castillo L, Martínez-Palacios CA (2002) Sunflower seed meal as a protein source in diets for *Tilapia rendalli* (Boulanger, 1896) fingerlings. *Aquaculture Research* **33**: 223–229.
- Orachunwong C, Thammasart S, Lohawatanakul C (2001) Recent developments in tilapia feeds. In: Proceedings of the Tilapia 2001 International Technical and Trade Conference on tilapia. INFOFISH, Kuala Lumpur, Malaysia, pp. 113–122.
- Pan Q, Liu S, Tan YG, Bi YZ (2003) The effect of chromium picolinate on growth and carbohydrate utilization in tilapia, *Oreochromis niloticus* × *Oreochromis aureus*. *Aquaculture* **225**: 421–429.
- Patel AB, Yakupitiyage A (2003) Mixed feeding schedules in semi-intensive pond culture of Nile tilapia, *Oreochromis niloticus*, L.: is it necessary to have two diets of different protein contents? *Aquaculture Research* **34**: 1343–1352.
- Peres H, Lim C, Klesius PH (2004) Growth, chemical composition and resistance to *Streptococcus iniae* challenge of Nile juvenile (*Oreochromis niloticus*) fed graded levels of dietary inositol. *Aquaculture* **235**: 423–432.
- Phelps RP, Okoko M (2011) A non-paradoxical dose response to 17 α -methyltestosterone by Nile tilapia *Oreochromis niloticus* (L.): effects on the sex ratio, growth and gonadal development. *Aquaculture Research* **42**: 549–558.
- Ponzoni RW, Nguyen NH, Khaw HL, Hamzah A, Bakar KRA, Yee HY (2011) Genetic improvement of Nile tilapia (*Oreochromis niloticus*) with special reference to the work conducted by the WorldFish Center with the GIFT strain. *Reviews in Aquaculture* **3**: 27–41.
- Rao DS, Raghuramulu N (1999) Vitamin D₃ and its metabolites have no role in calcium and phosphorus metabolism in *Tilapia mossambica*. *Journal of Nutritional Science and Vitaminology* **45**: 9–19.
- Reigh RC, Robinson EH, Brown PB (1991) Effects of dietary magnesium on growth and tissue magnesium content of blue tilapia *Oreochromis aureus*. *Journal of the World Aquaculture Society* **22**: 192–200.
- Riche M, Garling DL (2004) Effect of phytic acid on growth and nitrogen retention in tilapia *Oreochromis niloticus* (L.). *Aquaculture Nutrition* **10**: 389–400.
- Riche M, Trotter NL, Ku PK, Garling DL (2001) Apparent digestibility of crude protein and apparent availability of individual amino acids in tilapia (*Oreochromis niloticus*) fed phytase pretreated soybean meal diets. *Fish Physiology and Biochemistry* **25**: 181–194.
- Ridha MT (2006) A comparative study on the growth, feed conversion and production of fry of improved and non-improved strains of the Nile tilapia *Oreochromis niloticus*. *Asian Fisheries Science* **19**: 319–329.
- Robinson EH, Rawles SD, Oldenburg PW, Stickney RR (1984) Effects of feeding glandless or glanded cottonseed products and gossypol to *Tilapia aurea*. *Aquaculture* **28**: 145–154.
- Robinson EH, LaBomascus D, Brown PB, Linton TL (1987) Dietary calcium and phosphorus requirements of *Oreochromis aureus* reared in calcium-free water. *Aquaculture* **64**: 267–276.
- Rodríguez-Serna M, Olvera-Novoa MA, Carmona-Osalde C (1996) Nutritional value of animal by-product meal in practical diets for Nile tilapia *Oreochromis niloticus* (L.) fry. *Aquaculture Research* **27**: 67–73.
- Romana-Eguia MRR, Ikeda M, Basiao ZU, Taniguchi N (2010) Growth comparison of Asian Nile and red tilapia strains in controlled and uncontrolled farm conditions. *Aquaculture International* **18**: 1205–1221.
- Saha S, Roy RN, Sen SK, Ray AK (2006) Characterization of cellulase-producing bacterial from the digestive tract of tilapia, *Oreochromis mossambica* (Peters) and grass carp, *Ctenopharyngodon idella* (Valenciennes). *Aquaculture Research* **37**: 380–388.
- Saleh G, Eleraky W, Gropp JM (1995) A short note on the effects of vitamin A hypervitaminosis and hypovitaminosis on health and growth of *Tilapia nilotica* (*Oreochromis niloticus*). *Journal of Applied Ichthyology* **11**: 382–385.
- Santiago CB, Laron MA (2002) Growth and fry production of Nile tilapia, *Oreochromis niloticus* (L.), on different feeding schedules. *Aquaculture Research* **33**: 129–136.
- Santiago CB, Lovell R (1988) Amino acid requirements for growth of Nile tilapia. *Journal of Nutrition* **118**: 1540–1546.
- Santiago CB, Reyes OS (1993) Effects of dietary lipid source on reproductive performance and tissue lipid levels of Nile tilapia *Oreochromis niloticus* (Linnaeus) broodstock. *Journal of Applied Ichthyology* **9**: 33–40.
- Santiago CB, Aldaba MB, Abuan EF, Laron MA (1985) The effects of artificial diets on fry production and growth of *Oreochromis niloticus* breeders. *Aquaculture* **47**: 193–203.
- Santos LD, Furuya WM, da Silva LCR, Matsushita M, de Castro Silva TS (2011) Dietary conjugated linoleic acid (CLA) for finishing Nile tilapia. *Aquaculture Nutrition* **17**: e70–e81.
- Schulz C, Wickert M, Kijora C, Ogunji J, Rennert B (2007) Evaluation of pea protein isolate as alternative protein source in diets for juvenile tilapia (*Oreochromis niloticus*). *Aquaculture Research* **38**: 537–545.
- Sealey WM, Lim C, Klesius PH (1997) Influence of the dietary level of iron from iron methionine and iron sulfate on immune response and resistance of channel catfish to *Edwardsiella ictaluri*. *Journal of the World Aquaculture Society* **28**: 142–149.
- Shiau SY (1997) Utilization of carbohydrates in warmwater fish – with particular references to tilapia, *Oreochromis niloticus* × *O. aureus*. *Aquaculture* **151**: 79–96.
- Shiau SY (2002) Tilapia, *Oreochromis* spp. In: Webster CD, Lim C (eds) *Nutritional Requirements and Feeding of Finfish for Aquaculture*, pp. 273–292. CAB International, Wallingford.
- Shiau SY, Chen MJ (1993) Carbohydrate utilization by tilapia (*Oreochromis niloticus* × *O. aureus*) as influenced by different chromium sources. *Journal of Nutrition* **123**: 1747–1753.
- Shiau SY, Chin YH (1999) Estimation of dietary biotin requirement of juvenile hybrid tilapia, *Oreochromis niloticus* × *O. aureus*. *Aquaculture* **170**: 71–78.

- Shiau SY, Chuang JC (1995) Utilization of disaccharides by juvenile tilapia, *Oreochromis niloticus* × *O. aureus*. *Aquaculture* **133**: 249–256.
- Shiau SY, Hsieh HL (1997) Vitamin B-6 requirements of tilapia *Oreochromis niloticus* × *O. aureus* fed two dietary protein concentrations. *Fisheries Science* **63**: 1002–1007.
- Shiau SY, Hsieh JF (2001) Quantifying the dietary potassium requirement of juvenile hybrid tilapia (*Oreochromis niloticus* × *O. aureus*). *British Journal of Nutrition* **85**: 213–218.
- Shiau SY, Hsu TS (1995) L-Ascorbyl-2-sulfate has equal antiscorbutic activity as L-ascorbyl-2-monophosphate for tilapia, *Oreochromis niloticus* × *O. aureus*. *Aquaculture* **133**: 147–157.
- Shiau SY, Hsu TS (1999) Quantification of vitamin C requirement for juvenile hybrid, *Oreochromis niloticus* × *Oreochromis aureus*, with L-ascorbyl-2-monophosphate-Na and L-ascorbyl-2-monophosphate-Mg. *Aquaculture* **175**: 317–326.
- Shiau SY, Huang SY (2001) Dietary folic acid requirement for maximum growth of juvenile tilapia *Oreochromis niloticus* × *O. aureus*. *Fisheries Science* **67**: 655–659.
- Shiau SY, Hwang JY (1993) Vitamin D requirement of juvenile hybrid tilapia *Oreochromis niloticus* × *O. aureus*. *Nippon Suisan Gakkaishi* **59**: 553–558.
- Shiau SY, Liang HS (1995) Carbohydrate utilization and digestibility by tilapia, *Oreochromis niloticus* × *O. aureus*, are affected by chromic oxide inclusion in the diet. *Journal of Nutrition* **125**: 976–982.
- Shiau SY, Lin SF (1993) Effect of dietary chromium and vanadium on the utilization of different carbohydrates in tilapia, *Oreochromis niloticus* × *O. aureus*. *Aquaculture* **110**: 321–330.
- Shiau SY, Lo PS (2000) Dietary choline requirements of juvenile hybrid tilapia, *Oreochromis niloticus* × *O. aureus*. *Journal of Nutrition* **130**: 100–103.
- Shiau SY, Lu LS (2004) Dietary sodium requirement determined for juvenile hybrid tilapia (*Oreochromis niloticus* × *O. aureus*) reared in fresh water and seawater. *British Journal of Nutrition* **91**: 585–590.
- Shiau SY, Lung CQ (1993) No dietary vitamin B₁₂ required for juvenile tilapia *Oreochromis niloticus* × *O. aureus*. *Comparative Biochemistry and Physiology* **105A**: 147–150.
- Shiau SY, Peng CY (1993) Protein-sparing effect by carbohydrates in diets containing glucose or dextrin. *Journal of Nutrition* **122**: 2030–2036.
- Shiau SY, Shy SM (1998) Dietary chromic oxide inclusion level required to maximize glucose utilization in hybrid tilapia, *Oreochromis niloticus* × *O. aureus*. *Aquaculture* **161**: 357–364.
- Shiau SY, Su LW (2003) Ferric citrate is half as effective as ferrous sulfate in meeting the iron requirement of juvenile tilapia, *Oreochromis niloticus* × *O. aureus*. *Journal of Nutrition* **133**: 483–488.
- Shiau SY, Su SL (2005) Juvenile tilapia (*Oreochromis niloticus* × *O. aureus*) requires dietary myo-inositol for maximal growth. *Aquaculture* **243**: 273–277.
- Shiau SY, Suen GS (1992) Estimation of the niacin requirements for tilapia fed diets containing glucose or dextrin. *Journal of Nutrition* **122**: 2030–2036.
- Shiau SY, Tseng HC (2007) Dietary calcium requirements of juvenile tilapia, *Oreochromis niloticus* × *O. aureus*, reared in fresh water. *Aquaculture Nutrition* **13**: 298–303.
- Siddiqui AQ, Al-Harbi AH, Al-Hafedh YS (1997) Effects of food supply on size at first maturity, fecundity and growth of hybrid tilapia, *Oreochromis niloticus* (L.) × *Oreochromis aureus* (Steindachner), in outdoor concrete tanks in Saudi Arabia. *Aquaculture Research* **28**: 341–349.
- Siddiqui AQ, Al-Hafedh YS, Ali SA (1998) Effect of dietary protein level on the reproductive performance of Nile tilapia, *Oreochromis niloticus* (L.). *Aquaculture Research* **29**: 349–358.
- Sinha AK, Kumar V, Makkar HPS, Boeck GD, Becker K (2011) Non-starch polysaccharides and their role in fish nutrition – a review. *Food Chemistry* **127**: 1409–1426.
- Sintayehu A, Mathies E, Meyer-Burgdorff KH, Rosenow H, Günther KD (1996) Apparent digestibilities and growth experiments with tilapia (*Oreochromis niloticus*) fed soybean meal, cottonseed meal and sunflower seed meal. *Journal of Applied Ichthyology* **12**: 125–130.
- Soliman AK, Wilson RP (1992a) Water-soluble vitamin requirements of tilapia. 1. Pantothenic acid requirement of blue tilapia, *Oreochromis aureus*. *Aquaculture* **104**: 121–126.
- Soliman AK, Wilson RP (1992b) Water-soluble vitamin requirements of tilapia. 2. Riboflavin requirement of blue tilapia, *Oreochromis aureus*. *Aquaculture* **104**: 309–314.
- Soliman AK, Jauncey K, Roberts RJ (1994) Water-soluble vitamin requirements of tilapia: ascorbic acid (vitamin C) requirements of Nile tilapia, *Oreochromis niloticus* (L.). *Aquaculture and Fisheries Management* **25**: 269–278.
- Sørensen M, Penn M, El-Mowafi A, Storebakken T, Chunfang C, Øverland M *et al.* (2011) Effect of stachyose, raffinose and soya-saponins supplementation on nutrient digestibility, digestive enzymes, gut morphology and growth performance in Atlantic salmon (*Salmo salar*, L.). *Aquaculture* **314**: 145–152.
- Stickney RR, McGeachin RB (1983a) Responses of *Tilapia aurea* to semipurified diets of differing fatty acid composition. In: *Proceedings of the International Symposium on Tilapia in Aquaculture*, pp. 346–355. Tel Aviv University Press, Nazareth, Israel.
- Stickney RR, McGeachin RB (1983b) Effects of dietary lipid quality on growth and food conversion of tilapia. *Proceedings of the Annual Conference Southeastern Association of Fish and Wildlife Agencies* **37**: 352–357.
- Sugita H, Miyajima C, Deguchi Y (1991) The vitamin B₁₂-producing ability of the intestinal microflora of freshwater fish. *Aquaculture* **92**: 267–276.
- Sweilum MA, Abdella MM, Salah El-Din SA (2005) Effect of dietary protein-energy levels and fish initial sizes on growth rate, development and production of Nile tilapia, *Oreochromis niloticus* L. *Aquaculture Research* **36**: 1414–1421.
- Szabó A, Mézes M, Hancz C, Molnár T, Varga D, Romvári R *et al.* (2011) Incorporation dynamics of dietary vegetable oil fatty acids into the triacylglycerols and phospholipids of

- tilapia (*Oreochromis niloticus*) tissues (fillet, liver, visceral fat and gonads). *Aquaculture Nutrition* **17**: e132–e147.
- Tacon AGJ, De Silva SS (1983) Mineral composition of some commercial fish feeds available in Europe. *Aquaculture* **31**: 11–20.
- Tacon AGJ, Metian M (2008) Global overview on the use of fish meal and fish oil in industrially compounded aquafeeds: trends and future prospects. *Aquaculture* **285**: 146–158.
- Takeuchi T, Satoh S, Watanabe T (1983) Requirement of *Tilapia nilotica* for essential fatty acids. *Bulletin of the Japanese Society of Scientific Fisheries* **49**: 1127–1134.
- Tayamen MM, Shelton WL (1978) Inducement of sex reversal in *Sarotherodon niloticus* (Linnaeus). *Aquaculture* **14**: 349–354.
- Teoh CY, Turchini GM, Ng WK (2011) Genetically improved farmed Nile tilapia and red hybrid tilapia showed differences in fatty acid metabolism when fed diets with added fish oil or a vegetable oil blend. *Aquaculture* **312**: 126–136.
- Tocher DR, Angaba M, Hastings N, Bell JG, Dick R, Teale AJ (2002) Nutritional regulation of hepatic fatty acid desaturation and polyunsaturated fatty acid composition in zebrafish (*Danio rerio*) and tilapia (*Oreochromis niloticus*). *Fish Physiology and Biochemistry* **24**: 309–320.
- Tonial IB, Stevanato FB, Matsushita M, De Souza NE, Furuya WM, Visentainer JV (2009) Optimization of flaxseed oil feeding time length in adult Nile tilapia (*Oreochromis niloticus*) as a function of muscle omega-3 fatty acids composition. *Aquaculture Nutrition* **15**: 564–568.
- Trushenski JT, Boesenberg J, Kohler CC (2009) Influence of grow-out feed fatty acid composition on finishing success in Nile tilapia. *North American Journal of Aquaculture* **71**: 242–251.
- Tsadik GG, Bart AN (2007) Effects of feeding, stocking density and water-flow rate on fecundity, spawning frequency and egg quality of Nile tilapia, *Oreochromis niloticus* (L.). *Aquaculture* **272**: 380–388.
- Tung PS, Shiau SY (1993) Carbohydrate utilization versus body size in tilapia *Oreochromis niloticus* × *O. aureus*. *Comparative Biochemistry and Physiology* **104A**: 585–588.
- Turchini GM, Torstensen BE, Ng WK (2009) Fish oil replacement in finfish nutrition. *Reviews in Aquaculture* **1**: 10–57.
- Twibell RG, Brown PB (1998) Optimal dietary protein concentration for hybrid tilapia *Oreochromis niloticus* × *O. aureus* fed all-plant diets. *Journal of the World Aquaculture Society* **29**: 9–16.
- Viola S, Arieli Y, Zohar G (1988) Animal-protein-free feeds for hybrid tilapia (*Oreochromis niloticus* × *O. aureus*) in intensive culture. *Aquaculture* **75**: 115–125.
- Visentainer JV, de Souza NE, Makoto M, Hayashi C, Franco MRB (2005) Influence of diets enriched with flaxseed oil on the α -linolenic, eicosapentaenoic and docosahexaenoic fatty acid in Nile tilapia (*Oreochromis niloticus*). *Food Chemistry* **90**: 557–560.
- Wang Y, Yuen KH, Ng WK (2006) Deposition of tocotrienols and tocopherols in the tissues of red hybrid tilapia, *Oreochromis* sp., fed a tocotrienol-rich fraction extracted from crude palm oil and its effects on lipid peroxidation. *Aquaculture* **253**: 583–591.
- Weaver KL, Mester P, Chilton JA, Wilson MD, Pandey P, Chilton FH (2008) The content of favorable and unfavorable polyunsaturated fatty acids found in commonly eaten fish. *Journal of the American Dietetic Association* **108**: 1178–1185.
- Wedegaertner TC (1981) Making the most of cottonseed meal. *Feed Management Magazine* **32**: 13.
- Wilson RP (1994) Utilization of dietary carbohydrate by fish. *Aquaculture* **124**: 67–80.
- Woodward B (1994) Dietary vitamin requirements of cultured young fish, with emphasis on quantitative estimates for salmonids. A review. *Aquaculture* **124**: 133–168.
- Yigit NO, Olmez M (2011) Effects of cellulase addition to canola meal in tilapia (*Oreochromis niloticus* L.) diets. *Aquaculture Nutrition* **17**: e494–e500.
- Young K (2009) Omega-6 (n-6) and omega-3 (n-3) fatty acids in tilapia and human health: a review. *International Journal of Food Sciences and Nutrition* **60**: 203–211.
- Yue YR, Zhou QC (2008) Effect of replacing soybean meal with cottonseed meal on growth, feed utilization and hematological indexes for juvenile hybrid tilapia, *Oreochromis niloticus* × *O. aureus*. *Aquaculture* **284**: 185–189.
- Zhao H, Jiang R, Xue M, Xie S, Wu X, Guo L (2010) Fishmeal can be completely replaced by soy protein concentrate by increasing feeding frequency in Nile tilapia (*Oreochromis niloticus* GIFT strain) less than 2 g. *Aquaculture Nutrition* **16**: 648–653.
- Zhao HX, Cao JM, Liu XH, Zhu X, Chen SC, Lan HB *et al.* (2011) Effect of supplemental dietary zinc sources on the growth and carbohydrate utilization of tilapia Smith 1840, *Oreochromis niloticus* × *Oreochromis aureus*. *Aquaculture Nutrition* **17**: 64–72.
- Zhou QC, Yue YR (2010) Effect of replacing soybean meal with canola meal on growth, feed utilization and haematological indices of juvenile hybrid tilapia, *Oreochromis niloticus* × *Oreochromis aureus*. *Aquaculture Research* **41**: 982–990.
- Zhou QC, Yue YR (2012) Apparent digestibility coefficients of selected feed ingredients for juvenile hybrid tilapia, *Oreochromis niloticus* × *Oreochromis aureus*. *Aquaculture Research* **43**: 806–814.